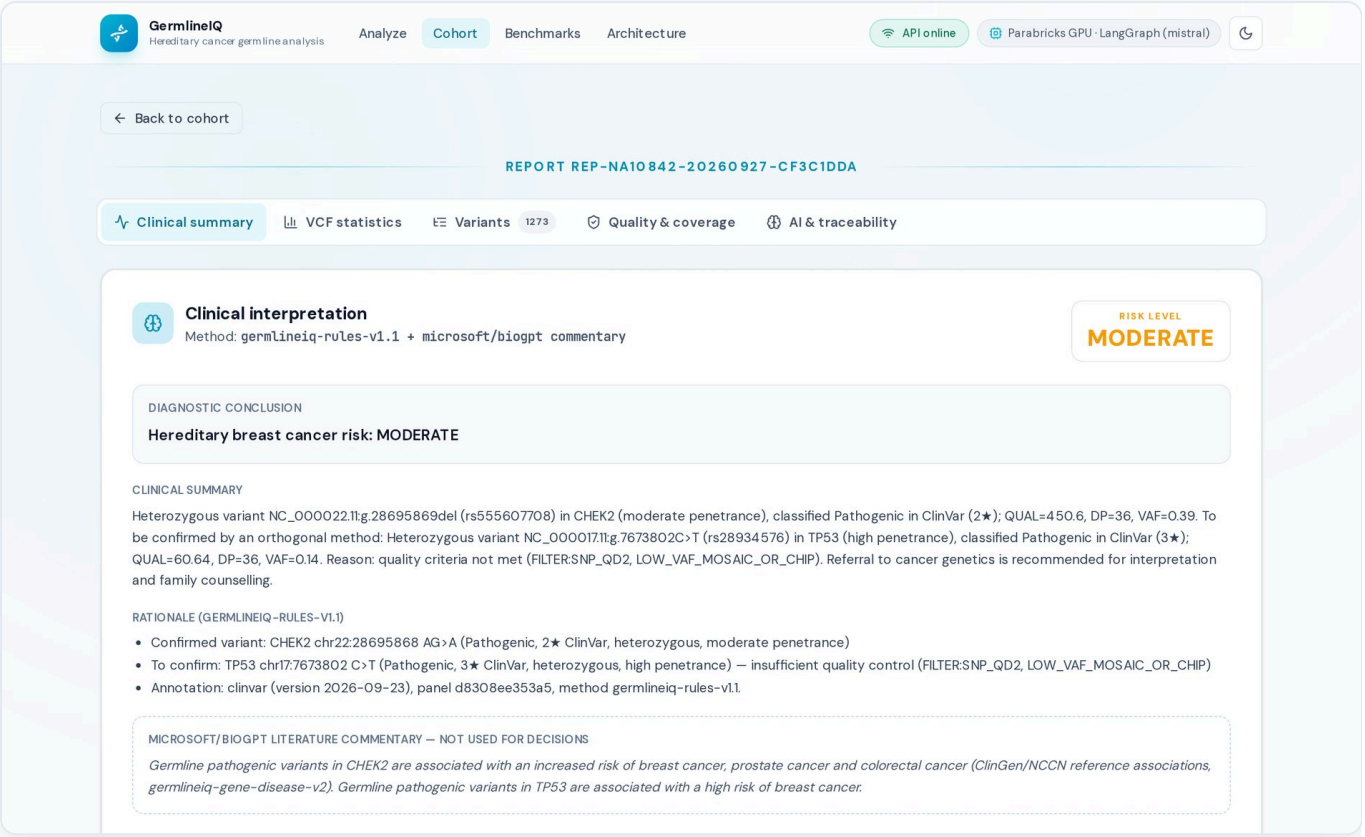
Demonstration & Validation Report

Multi-agent clinical genomics platform for hereditary breast cancer risk — from FASTQ reads to a verified clinical report on a local GPU server



Software version	GermlineIQ 3.0.0 · commit 83cc9a4
Report date	27 September 2026
Validation server	NVIDIA GeForce RTX 5080 (16 GB) · 6 CPUs · 18 GB RAM · WSL2
Pipeline in service	NVIDIA Parabricks 4.6.0-1 (GPU) with GATK 4.2.6.1 · LangGraph (mistral)
Reference data	GRCh38 (GATK resource bundle) · ClinVar GRCh38 release 2026-09-23 · GIAB HG001 v4.2.1 · 1000 Genomes 30× (NYGC)

Research software. Results are decision support only, not a medical diagnosis, and must be validated by a clinical geneticist or an oncologist.

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> **How to read this report.** Every screenshot was taken automatically from the running system (Chromium, 1440 px, light theme) on the validation server; every number in tables and text is read from the system's own output files (clinical reports, `vcf_statistics.json`, benchmark results, model evaluation) or from the independent verification script, never typed by hand.

1. Executive summary

GermlineIQ turns paired-end sequencing reads (FASTQ) or a variant file (VCF) into a hereditary breast cancer risk report for a panel of 13 germline genes (BRCA1, BRCA2, PALB2, TP53, PTEN, CDH1, STK11 — high penetrance; CHEK2, ATM, BARD1, RAD51C, RAD51D, NF1 — moderate penetrance). A LangGraph orchestrator plans and runs six agents: data preparation, GPU variant calling (NVIDIA Parabricks, with GATK4 on CPU as fallback), ClinVar annotation, quality control and statistics, clinical interpretation, and report assembly. Everything runs on a local server: health data never leave the machine.

The risk level is decided by explicit, versioned rules (`germlineiq-rules-v1.1`), never by a language model. BioGPT contributes two texts — a literature commentary and, after fine-tuning on VCF statistics, an interpretation of each analysis — and **every sentence is checked by a deterministic verifier** before it can reach the report; unsupported sentences are replaced by reference text.

Key biological result. On real genomes the system found every inherited loss-of-function variant — two BRCA2 truncations, a PALB2 frameshift, the CHEK2 c.1100delC founder variant (whose paternal transmission is confirmed by the 1000 Genomes trio) and an ATM splice variant — and it refused to call a TP53 p.Arg273His hotspot present in only 5 of 36 reads: an acquired event which, treated as germline, would have produced a false Li-Fraumeni diagnosis (chapter 8).

100 %
Risk-level accuracy
7/7 real 1000 Genomes genomes · 95 % CI 65–100 %

0
Unexpected confirmed P/LP calls
1 low-VAF somatic-like call correctly flagged “to confirm”

97.5 %
GIAB precision (PASS)
recall 88.8 % · all calls 100 %

20/20
Independent verification
every report metric recomputed with bcftools/samtools

1,357/1,357
GPU = CPU
identical final VCF (Parabricks vs GATK4)

100 %
BioGPT sentences verified
15 held-out real analyses · 98.6 % on 300 synthetic

identical
Reproducibility
same input → same report (SHA-256)

9/9
Fault-injection cases
service healthy afterwards

Table 1. Key results of the validation (final benchmark on the delivery server, GPU pipeline).

Dimension	Result	Evidence
Clinical accuracy	Carrier sensitivity 100 % (5/5), control specificity 100 % (2/2), risk level 7/7	Ch. 6 — clinical accuracy
Analytical validity	PASS calls: precision 97.5 %, recall 88.8 %, F1 0.929; all calls: recall 100 %	Ch. 6 — analytical validity
Verified AI	Statistics interpreter: 98.6 % (synthetic) / 100 % (real) verified sentences, 0 wrong risk levels; base BioGPT: 0 %	Ch. 6 — verified AI
Determinism	Identical reports and statistics over repeated runs (VCF and FASTQ); deterministic and Mistral routers give the same report	Ch. 6 — determinism
Robustness & latency	9/9 fault cases handled; read-endpoint p95 < 1.5 ms	Ch. 6 — robustness
Speed	FASTQ → report median 99 s per sample (≈150k read pairs, GPU)	Ch. 6 — clinical accuracy

2. Biological background and the problem addressed

Why hereditary breast cancer is a DNA-repair disease

Every human cell suffers DNA double-strand breaks — from replication stress, reactive oxygen species or ionising radiation. Two families of proteins respond. **Signalling and checkpoint proteins** detect the break and stop the cell cycle: ATM is recruited by the MRN complex and phosphorylates CHEK2 and p53 (TP53); p53 then arrests the cycle or triggers apoptosis if the damage is too severe. **Repair proteins** fix the break without error by *homologous recombination* (HR), which copies the intact sister chromatid: BRCA1 (with its partner BARD1) promotes resection of the broken ends, PALB2 bridges BRCA1 to BRCA2, and BRCA2 loads the recombinase RAD51 — helped by the paralogues RAD51C and RAD51D — onto single-stranded DNA for strand invasion [Moynahan & Jasin 2010; Xia et al. 2006].

When HR fails, cells fall back on error-prone pathways (non-homologous end joining), accumulate rearrangements and mutations, and can progress towards cancer. Nine of the thirteen genes of the GermlineIQ panel act directly in this response (Figure 1); the four others (PTEN, STK11, CDH1, NF1) are tumour suppressors of growth-signalling and cell-adhesion pathways whose germline loss also raises breast cancer risk.

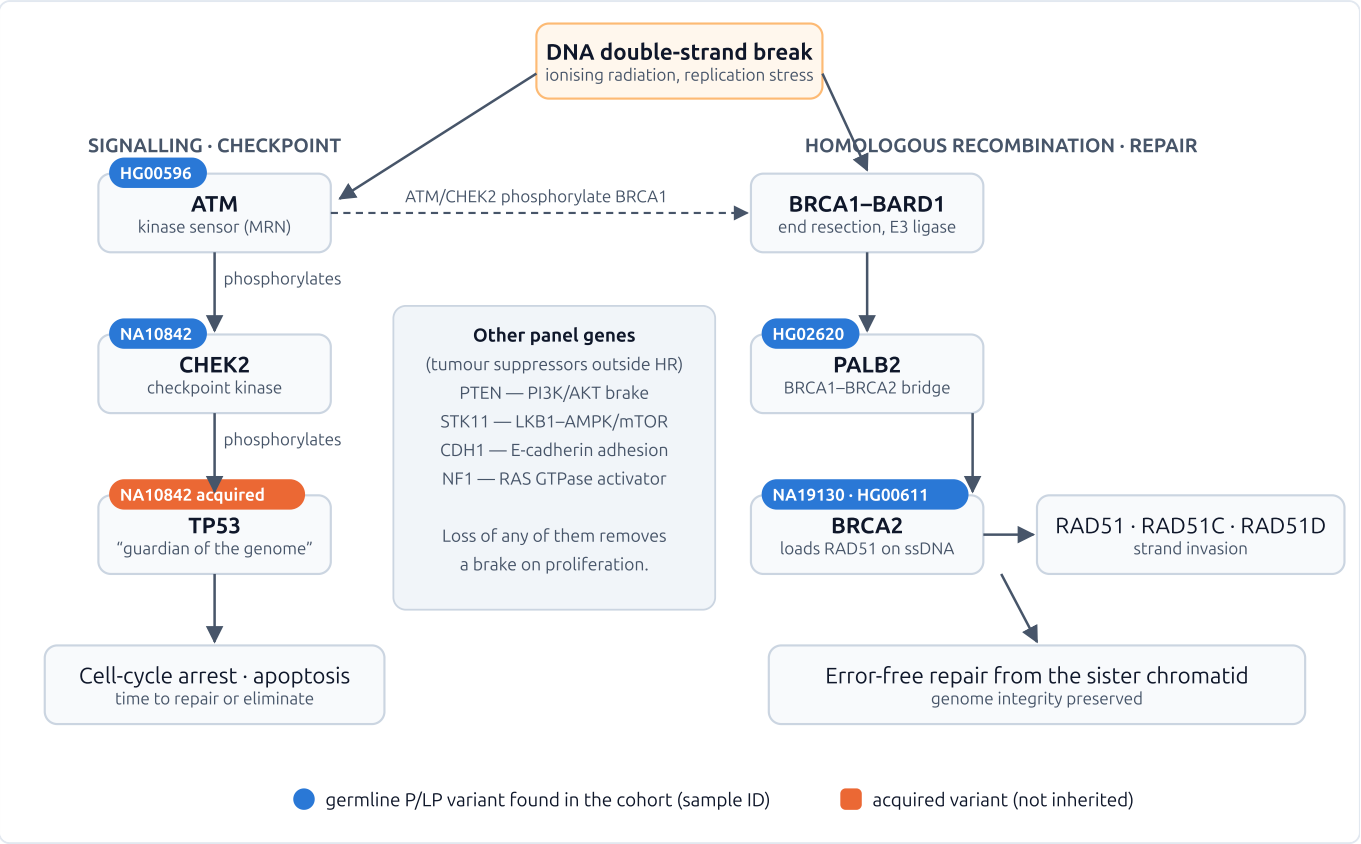


Figure 1. DNA double-strand-break response and homologous recombination, with the panel genes. Badges mark the genes in which the validation cohort carries a pathogenic variant (blue: germline, inherited; orange: acquired, not inherited).

From one inherited variant to a cancer: the two-hit model

A person who inherits a **heterozygous** loss-of-function variant in one of these genes carries it in every cell, and still has one working copy. A tumour arises when the second copy is lost in a breast cell (a somatic "second hit", often by loss of heterozygosity) — Knudson's two-hit model [Knudson 1971]. Inheritance is **autosomal dominant**: each child of a carrier has a 50 % probability of inheriting the variant, whatever the parent's sex. The probability that a carrier develops breast cancer — the **penetrance** — depends on the gene:

Table 2. The 13 germline genes of the panel: biology and breast cancer risk (OR: odds ratio for protein-truncating variants, BCAC 2021).

Gene	Biological role	Associated conditions	Breast cancer risk of a pathogenic variant	Class
BRCA1	HR initiation with BARD1; checkpoint control	Hereditary breast and ovarian cancer	≈ 72 % cumulative risk to age 80 [Kuchenbaecker 2017]; OR ≈ 10.6 [BCAC 2021]	High
BRCA2	Loads RAD51 onto single-stranded DNA	HBOC; pancreatic, prostate, male breast cancer; biallelic: Fanconi anaemia	≈ 69 % to age 80 [Kuchenbaecker 2017]; OR ≈ 5.9 [BCAC 2021]	High
PALB2	Partner and localiser of BRCA2	Breast, pancreatic, ovarian cancer; biallelic: Fanconi anaemia	OR ≈ 5.0 [BCAC 2021]; ≈ 35 % to age 70 [Antoniou 2014]	High
TP53	Transcription factor: arrest, apoptosis, senescence	Li-Fraumeni syndrome (early-onset breast cancer, sarcoma, brain tumours)	Very high, early onset [Malkin 1990]	High
PTEN · CDH1 · STK11	PI3K/AKT brake · E-cadherin adhesion · LKB1–AMPK/mTOR	PTEN hamartoma tumour syndrome · diffuse gastric and lobular breast cancer · Peutz-Jeghers syndrome	High (syndromic)	High
CHEK2	Checkpoint kinase phosphorylated by ATM	Breast, prostate, colorectal cancer	OR ≈ 2.5 [BCAC 2021]; c.1100delC OR ≈ 2.3 [CHEK2 consortium 2004]	Moderate
ATM	Apical DNA-damage kinase	Breast, pancreatic, prostate cancer; biallelic: ataxia-telangiectasia	OR ≈ 2.1 [BCAC 2021; Renwick 2006]	Moderate
BARD1 · RAD51C · RAD51D	BRCA1 partner · RAD51 paralogues	Breast (BARD1); ovarian and breast (RAD51C/D)	Moderate [BCAC 2021; Hu 2021]	Moderate
NF1	RAS GTPase-activating protein	Neurofibromatosis type 1	Moderate, mainly before age 50	Moderate

What makes a variant pathogenic

Most of the ≈1,300 variants that a person carries in these genes are harmless: common polymorphisms, intronic changes, synonymous substitutions. A pathogenic variant usually **destroys the protein's function**:

- **Nonsense** — a premature stop codon; the mRNA is degraded by nonsense-mediated decay or produces a truncated protein [Lykke-Andersen & Jensen 2015].
- **Frameshift** — an insertion or deletion whose length is not a multiple of three shifts the reading frame and soon creates a stop codon.
- **Canonical splice site** — the change of a GT/AG splice signal causes exon skipping or intron retention.
- **Missense in a critical residue** — a single amino-acid change in an essential domain, e.g. the DNA-contact residues of the p53 DNA-binding domain [Cho et al. 1994].

ClinVar aggregates the classifications submitted by laboratories; its review status (from one star to four) measures the strength of the evidence, "reviewed by expert panel" being the highest level routinely reached [Landrum 2018; Richards 2015].

What sequencing actually measures

The human genome is **diploid**: at a heterozygous germline site, half of the DNA molecules carry the variant, so about 50 % of the reads should show it (variant allele fraction, VAF ≈ 0.5), and ≈ 100 % at a homozygous site. The read count follows a binomial law, so at 30× the VAF of a true heterozygote fluctuates roughly between 0.3 and 0.7. A variant present in only a fraction of the cells — a **somatic** or **mosaic** event, clonal haematopoiesis in blood, or a mutation acquired by a cell line in culture — gives a clearly lower VAF [Jaiswal 2014; Genovese 2014]. Two genome-wide ratios also have a biological basis: transitions (A↔G, C↔T) outnumber transversions because methylated cytosines in CpG dinucleotides spontaneously deaminate into thymine, which gives Ti/Tv ≈ 2.0–2.1 genome-wide and ≈ 3 in coding exons [DePristo 2011]; and the heterozygous/homozygous ratio reflects population history — genomes of African ancestry carry more heterozygous sites than genomes whose ancestors went through the out-of-Africa bottleneck [1000 Genomes 2015].

The problem and the solution provided by GermlineIQ

Turning sequencing reads into a reliable hereditary-risk statement raises biological and technical problems that a variant caller alone does not solve. The table maps each problem to the mechanism GermlineIQ implements and to the evidence observed on real data in this report.

Table 3. From problem to solution.

Biological / clinical problem	Mechanism in GermlineIQ	Evidence on real data
A needle in a haystack. A pathogenic variant is one among ≈1,300 variants in the panel genes.	Exact-allele ClinVar matching after normalisation; classification of every variant; P/LP restricted to germline panel genes.	5/5 carriers found, 0 false confirmed P/LP among 10,639 panel variants of the seven genomes.
Penetrance differs by gene. A CHEK2 carrier is not a BRCA1 carrier.	Rules by penetrance class: HIGH vs MODERATE; knowledge base checks every risk magnitude written by BioGPT.	7/7 risk levels correct; BioGPT's "high risk" for CHEK2 rejected.
Somatic, mosaic or culture-acquired variants mimic germline ones.	VAF-aware clinical QC (heterozygous 0.25–0.75): a low-VAF P/LP call is reported "to confirm" and excluded from the risk.	TP53 p.Arg273His (5/36 reads) in NA10842 excluded — absent from both parents: acquired. Without this gate the report would have said HIGH (Li-Fraumeni).
A negative result is only as good as the coverage. Pseudogenes (CHEK2) and GC-rich or off-target exons (STK11, TP53 in exomes) are poorly covered.	Depth measured at each of the 26,090 known pathogenic sites; < 90 % → INDETERMINATE; any gene < 95 % named "not excluded".	CHEK2 93.9 % in NA10842; STK11 90.8 % and TP53 92.2 % in the NA12878 exome — stated, not hidden.

Biological / clinical problem	Mechanism in GermlineIQ	Evidence on real data
Sequencing artefacts (strand bias, low mapping quality, low depth).	GATK hard filters, MAPQ/base-quality thresholds; a filtered P/LP variant is never discarded but reported “to confirm”.	GIAB precision 97.5 % (PASS) with 100 % recall on all calls.
Clinicians need explanations, and language models hallucinate.	BioGPT fine-tuned on VCF statistics; every sentence verified against the computed values; reference fallback.	100 % verified sentences on 15 held-out real analyses; 0 wrong risk levels.
Clinical results must be reproducible and auditable.	Versioned rules, SHA-256 of inputs and content, JSON artifact per agent, independent recomputation.	Identical reports over repeated runs; 20/20 checks on 8/8 samples.
Genetic data are sensitive; turnaround matters.	Local GPU server, no cloud; Parabricks with automatic GATK fallback.	FASTQ → report in ≈ 99 s per sample; GPU and CPU calls identical.

3. System overview and architecture

What the system does

Table 4. Two entry points; the orchestrator computes the plan from the data provided.

Input	Agent chain	Output
FASTQ R1 + R2	DataManager → VariantCalling (Parabricks GPU or GATK4 CPU) + alignment QC → VariantAnnotation (ClinVar) → VCFAnalysis (QC, classification, statistics) → Prediction (rules + verified BioGPT) → ReportGenerator	Clinical JSON report + archived artifacts
VCF	VariantAnnotation → VCFAnalysis → Prediction → ReportGenerator (alignment skipped)	Clinical JSON report

Design principles

- **Deterministic decision.** The risk level comes from explicit rules; same input, panel and ClinVar release → same report. The SHA-256 of the input and every version (panel, ClinVar, rules, statistics, verifier) are written into the report.
- **Conservative.** A pathogenic variant failing quality thresholds is reported "to confirm" (risk INDETERMINATE); insufficient coverage of known pathogenic sites never yields a false "low risk".
- **Verified AI.** Language models write text, never decisions; each generated sentence is checked against reference knowledge or against the computed statistics.
- **Auditable.** Every agent writes a sorted JSON artifact in `patients/<ID>/output/`; an independent script recomputes every metric with reference tools.
- **Local.** Services bound to 127.0.0.1; data under a single root on the server disk; public reference data only.

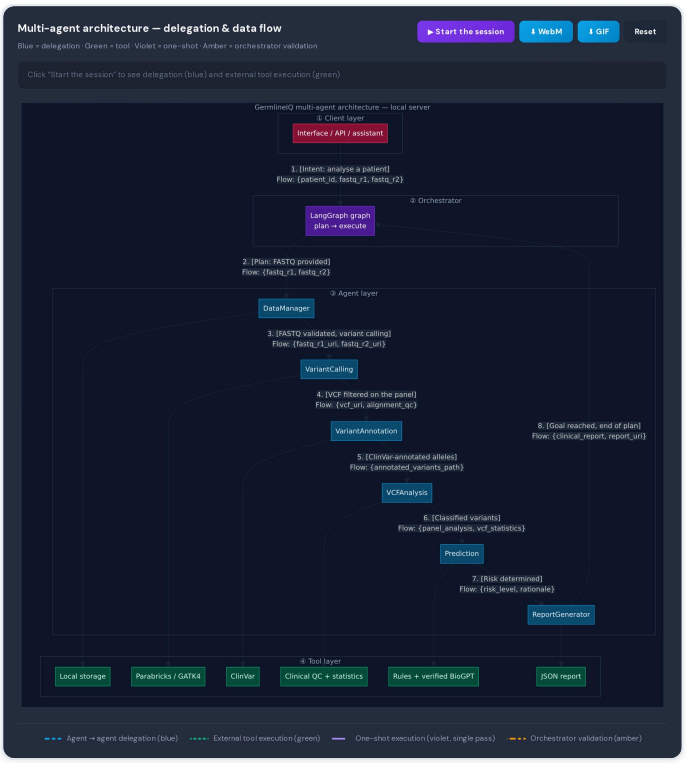


Figure 2. Multi-agent architecture as shown in the *Architecture* page: client layer (interface, API, assistant), LangGraph orchestrator (plan ⇄ execute), the six agents and the tools they call. Arrows carry the context keys exchanged between agents (e.g. `vcf_uri`, `panel_analysis`, `risk_level`).

4. Interface walkthrough

The web interface (Next.js) has four pages — **Analyze**, **Cohort**, **Benchmarks**, **Architecture** — and every clinical report has five tabs. The header permanently shows the API status and the engine selected on this server ("Parabricks GPU · LangGraph (mistral)").

Analyze page: starting an analysis

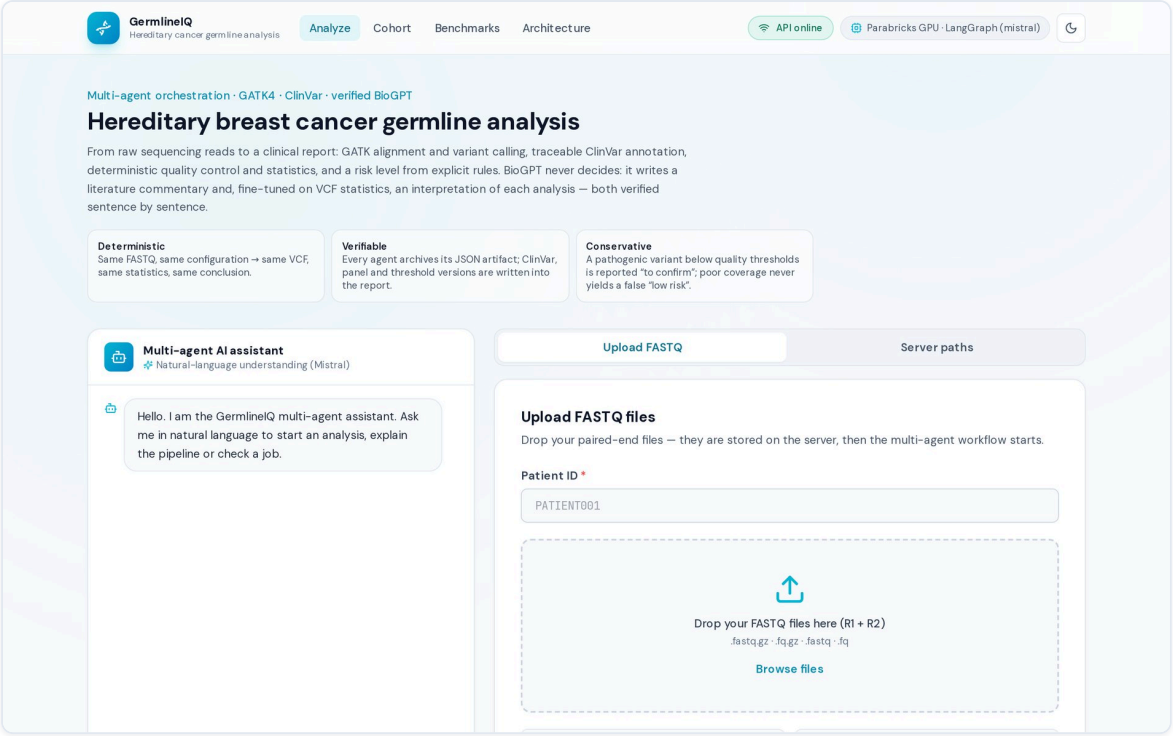


Figure 3. Analyze page. Left: the multi-agent assistant (Mistral v0.3 through Ollama) accepts natural-language requests. Right: FASTQ upload (drag and drop) or server paths. The three principle cards summarise the guarantees of the system.

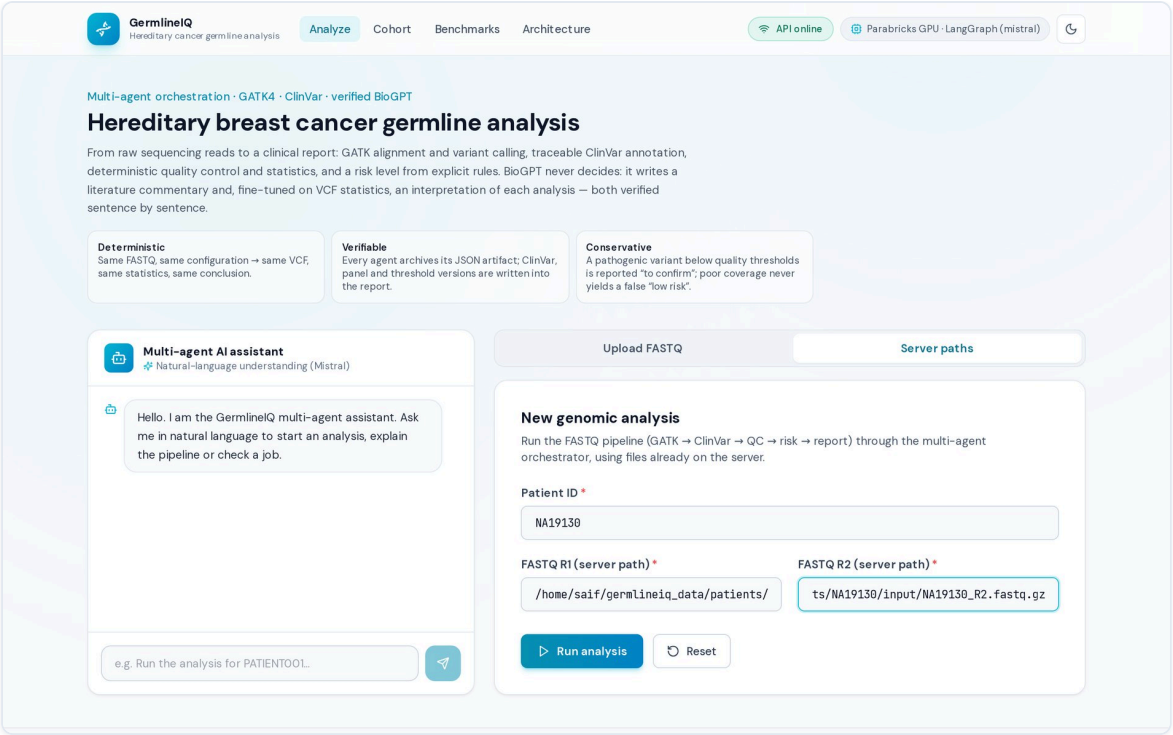


Figure 4. Server-paths form filled with the real 1000 Genomes sample NA19130 (BRCA2 carrier). Paths are validated by the API: they must exist, lie under the data root, and R1 must differ from R2.

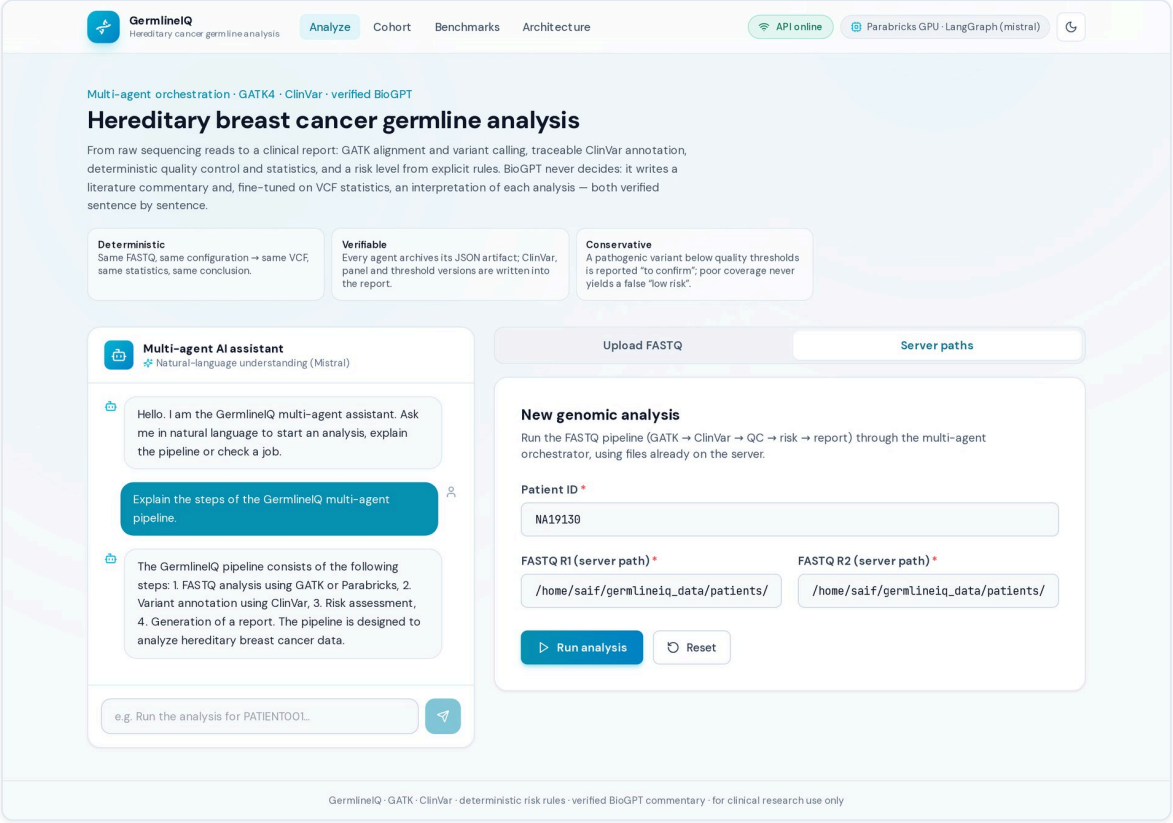


Figure 5. The assistant answering a question about the pipeline. Its intent parser recognises three actions — start a FASTQ analysis, start a VCF analysis, report a job status — and asks for missing fields.

Live execution tracking

The analysis below was started by typing "*Analyze the VCF …/patient_brca2_lowvaf.vcf for patient REPORT_DEMO*" in the assistant. The tracking card refreshes every four seconds from `GET /api/v1/jobs/<id>`.

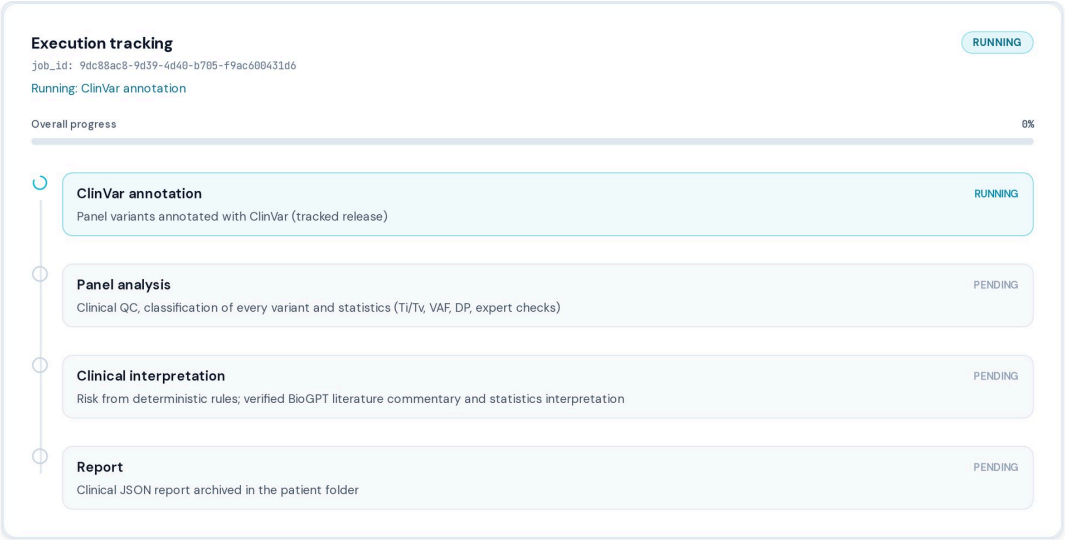


Figure 6. Seconds after submission: ClinVar annotation running, the other steps pending.

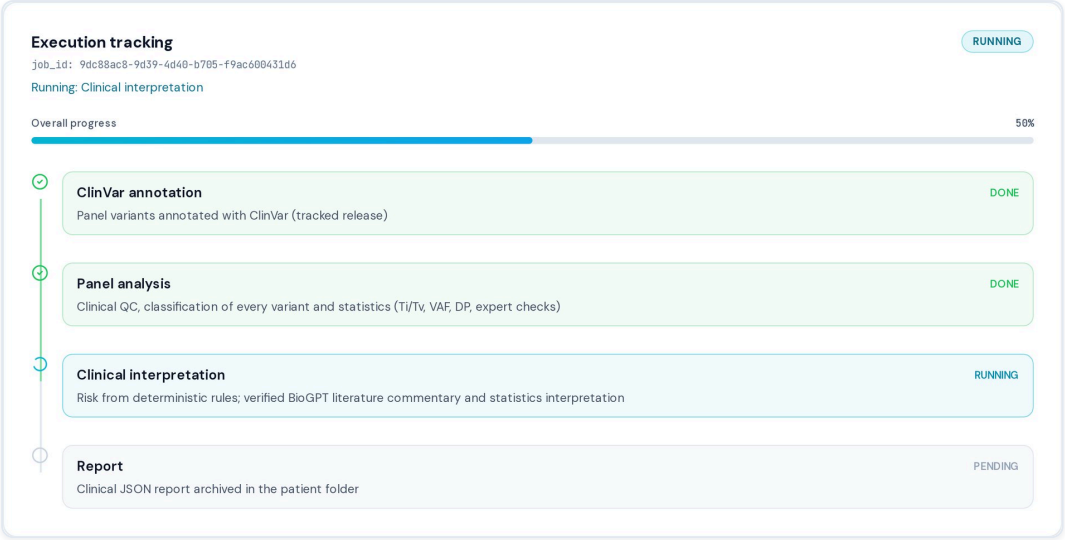


Figure 7. Annotation and panel analysis done; clinical interpretation (rules + BioGPT) running, progress 50 %.

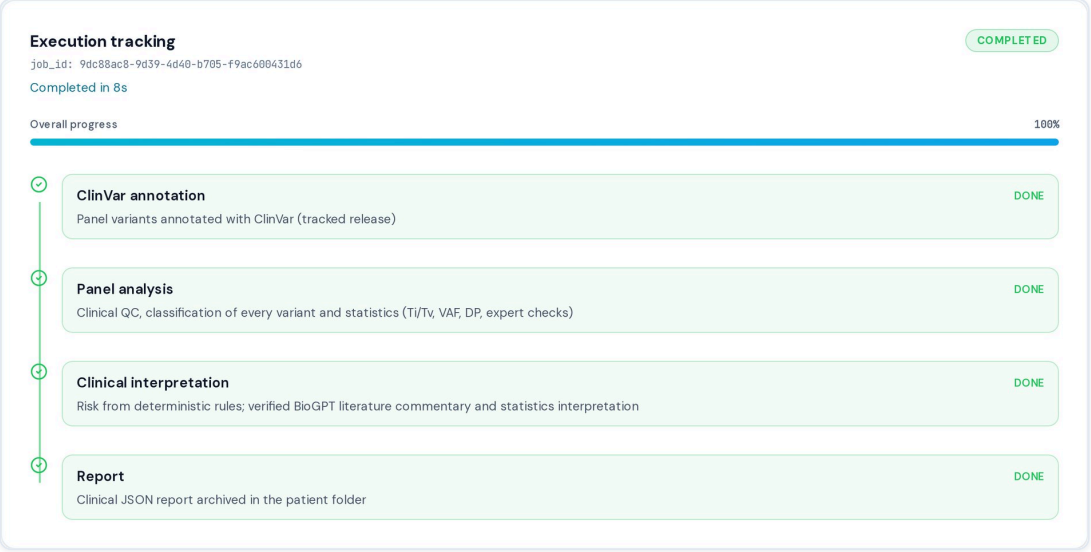


Figure 8. Completed job (VCF mode: the orchestrator skipped alignment). The clinical report opens below the tracking card.

The clinical report — NA10842 (CHEK2 carrier, 1000 Genomes)

NA10842 is a real 1000 Genomes genome carrying the CHEK2 c.1100delC founder variant. It is the most instructive case: one confirmed moderate-penetrance variant, one high-penetrance TP53 call that fails quality control, a gene not fully covered, and a BioGPT sentence rejected by the verifier.

Clinical interpretation
Method: `germlineiq-rules-v1.1 + microsoft/biogpt commentary`

RISK LEVEL
MODERATE

DIAGNOSTIC CONCLUSION

Hereditary breast cancer risk: MODERATE

CLINICAL SUMMARY

Heterozygous variant NC_000022.11:g.28695869del (rs555607708) in CHEK2 (moderate penetrance), classified Pathogenic in ClinVar (2★); QUAL=450.6, DP=36, VAF=0.39. To be confirmed by an orthogonal method: Heterozygous variant NC_000017.11:g.7673802C>T (rs28934576) in TP53 (high penetrance), classified Pathogenic in ClinVar (3★); QUAL=60.64, DP=36, VAF=0.14. Reason: quality criteria not met (FILTER:SNP_QD2, LOW_VAF_MOSAIC_OR_CHIP). Referral to cancer genetics is recommended for interpretation and family counselling.

RATIONALE (GERMLINEIQ-RULES-V1.1)

- Confirmed variant: CHEK2 chr22:28695868 AG>A (Pathogenic, 2★ ClinVar, heterozygous, moderate penetrance)
- To confirm: TP53 chr17:7673802 C>T (Pathogenic, 3★ ClinVar, heterozygous, high penetrance) — insufficient quality control (FILTER:SNP_QD2, LOW_VAF_MOSAIC_OR_CHIP)
- Annotation: clinvar (version 2026-09-23), panel d8308ee353a5, method germlineiq-rules-v1.1.

MICROSOFT/BIOGPT LITERATURE COMMENTARY — NOT USED FOR DECISIONS

Germline pathogenic variants in CHEK2 are associated with an increased risk of breast cancer, prostate cancer and colorectal cancer (ClinGen/NCCN reference associations, germlineiq-gene-disease-v2). Germline pathogenic variants in TP53 are associated with a high risk of breast cancer.

▸ ANALYSIS LIMITATIONS

MEDICAL DISCLAIMER

This report is generated by a decision-support system. It is not a medical diagnosis and must be validated by an oncologist or a clinical geneticist before any clinical or therapeutic decision.

`Status: AWAITING_MEDICAL_VALIDATION`

Figure 9. Tab *Clinical summary* — clinical interpretation card: risk level MODERATE from the deterministic rules, conclusion, rationale (confirmed CHEK2 variant; TP53 to confirm) and limitations.

Interpretation of the VCF statistics

Final text verified

Written by BioGPT fine-tuned on VCF statistics. Every sentence was checked against the computed statistics (numbers, expected ranges, risk level, genes); topics without a verified sentence use the deterministic reference text.

Generated sentences	Verified	Rejected	Topics from the model
14	14	0	14/14

BioGPT · verified VARIANT CALLS	1273 variants were called in the panel genes: 960 SNVs and 313 indels.
BioGPT · verified GATK FILTERS	98.1% of the calls passed the GATK filters, as expected for good-quality data.
BioGPT · verified TI/TV	The transition/transversion ratio (2.04) is within the expected range.
BioGPT · verified HET/HOM	The het/hom ratio (2.12) is within the expected range.
BioGPT · verified HETEROZYGOUS VAF	The median heterozygous VAF (0.50) is within the expected range.
BioGPT · verified DEPTH	Variant sites have a median depth of 30x, adequate for clinical calling.
BioGPT · verified MAPPING	99.9% of reads mapped to hg38, within the expected range (≥ 95%).
BioGPT · verified DUPLICATES	Duplicates account for 7.2% of read pairs, which is acceptable.
BioGPT · verified PATHOGENIC-SITE COVERAGE	99.7% of known pathogenic sites are covered at 15x or more, as expected.
BioGPT · verified NOT EXCLUDED	A known pathogenic variant cannot be excluded in CHEK2 because of incomplete coverage.
BioGPT · verified CONFIRMED P/LP	A pathogenic or likely pathogenic variant was confirmed in CHEK2.
BioGPT · verified TO CONFIRM	Variants in TP53 require orthogonal confirmation.
BioGPT · verified LOW-VAF CALLS	11 calls have a low allele fraction (possible mosaicism or clonal haematopoiesis) and are flagged for review.
BioGPT · verified RISK LEVEL	Hereditary breast cancer risk: MODERATE (deterministic rules).

Verification details (0 rejected sentences)

Figure 10. Interpretation of the VCF statistics written by BioGPT fine-tuned on statistics. Each sentence carries its source badge; here all 14 sentences were produced by the model and all passed the deterministic verifier (numbers, expected ranges, genes, risk level).

From FASTQ to report

What each orchestrator agent received and produced for this patient.

GENOMIC_PIPELINE	HAPLOTYPECALLER	VARIANT_ANNOTATION	VCF_ANALYSIS	VCF_ANALYSIS	PREDICTION
Sequenced reads	Alleles in the VCF	Panel variants	Clinical QC passed	Pathogenic / to confirm	Risk level
286.9k	1,273	1,273	1,159	2	MODERATE
99.9% mapped to hg38 · 7.2% duplicates	98.1% PASS · 1347 records read	21.3% ClinVar-annotated 2026-09-23	91.0% · QUAL, DP, VAF, FILTER	0 VUS · 0 conflicting	germlineiq-rules-v1.1

Figure 11. From FASTQ to report: the data journey — reads, alignment (mapping and duplicate rates), variant calls restricted to the panel, ClinVar annotation and classification.

Agent execution timeline

Wall time 7.5 s · agents 7.3 s · orchestration overhead 154 ms · router mistral

Table

Data preparation	1 ms
Variant calling	cached result
ClinVar annotation	261 ms
Panel analysis	43 ms
Interpretation	7.0 s
Report	23 ms

Figure 12. Agent execution timeline with per-agent durations and the router used. For this run variant calling was reused from the result cache (same FASTQ files, same configuration), hence the short total time.

Confirmed pathogenic variants

Identified genes: CHEK2

GENE	MUTATION	LOCUS	QUAL	DP	VAF	ZYGOSITY	CLINVAR
CHEK2	AG>A rs555607708	chr22:28695868	450.60	36	38.9%	het	Pathogenic ★★☆☆

Figure 13. Confirmed pathogenic variant: CHEK2 NC_000022.11:g.28695869del (rs555607708, c.1100delC), heterozygous, ClinVar Pathogenic 2★, with GATK metrics.

Variants to confirm (orthogonal method)

GENE	MUTATION	LOCUS	QUAL	DP	VAF	ZYGOSITY	CLINVAR	REASON
TP53	C>T rs28934576	chr17:7673802	60.60	36	13.9%	het	Pathogenic ★★★☆	FILTER:SNP_QD2, LOW_VAF_MOSAIC_OR_CHIP

Figure 14. Variant to confirm: TP53 NC_000017.11:g.7673802C>T (rs28934576, p.Arg273His), ClinVar Pathogenic 3★ (expert panel) — but VAF 0.14 and a GATK filter (QD < 2): reported for orthogonal confirmation, not as a germline finding.

Variant explorer

Every germline panel variant, with its ClinVar classification and clinical quality control.

Benign / likely benign (269) | VUS / conflicting (0) | Not in ClinVar / other (1002) | Pathogenic / to confirm (2) | All genes | Clinical QC failed only

HGVS, rsID, condition, position...

Showing 1273 of 1273 variants

Classification	Gene	Position (hg38)	Alleles	GT	QUAL	DP	VAF	FILTER / QC	ClinVar
Pathogenic — to confirm	TP53	chr17:7,673,802	C>T SNV	0/1	60.6	36	0.139	SNP_QD2 FILTER:SNP_QD2 · LOW_VAF_MOSAIC_OR_CHIP	Pathogenic ★★★☆ rs28934576 · VCV 12366 — TP53-related discord...
Pathogenic — confirmed	CHEK2	chr22:28,695,868	AG>A Deletion	0/1	450.6	36	0.389	PASS	Pathogenic ★★☆☆ rs555607708 · VCV 128042 — TUMOR PREDISPO...
Not in ClinVar	STK11	chr19:1,178,520	C>CT Insertion	1/1	494.0	23	1.000	PASS	not listed
Not in ClinVar	STK11	chr19:1,178,627	A>C SNV	1/1	1046.1	30	1.000	PASS	not listed
Not in ClinVar	STK11	chr19:1,178,956	C>T SNV	1/1	2059.1	56	1.000	PASS	not listed
Not in ClinVar	STK11	chr19:1,179,375	C>T SNV	1/1	1597.1	43	1.000	PASS	not listed
Not in ClinVar	STK11	chr19:1,180,010	CTT>C Deletion	0/1	215.6	16	0.375	PASS	not listed
Not in ClinVar	STK11	chr19:1,180,014	CTTTCTTTCTTTTCTTTCTCTCTTTCTT>C Deletion	0/1	214.6	16	0.375	PASS	not listed
Not in ClinVar	STK11	chr19:1,180,839	GTCTGTCTCTGTCTCTGTCTCTA>G Deletion	1/1	1584.0	36	1.000	PASS	not listed

Figure 15. Tab *Variants* — variant explorer over the 1,273 panel variants: search, filters by category and QC status, per-variant GATK metrics and ClinVar classification.

Expert quality checks

8 pass · 0 warning(s) · 0 not assessable

Each metric is compared with the range expected for a human germline sample. A warning flags data to review; only coverage of known pathogenic sites enters the risk rules.

Check	Value	Expected	Status	Interpretation
Transition/transversion ratio (PASS SNVs)	2.04	1.8–3.3 (≥ 30 SNVs)	Pass	A low Ti/Tv indicates false positives (random errors ≈ 0.5); ≈ 2.0 genome-wide, ≈ 3.0 coding exome.
Heterozygous/homozygous ratio	2.12	1.0–3.0 (≥ 30 variants)	Pass	Too high: possible contamination; too low: consanguinity or loss of heterozygosity.
Median heterozygous VAF	0.500	0.4–0.6 (≥ 10 heterozygotes)	Pass	A germline heterozygote is expected near 0.5; a shift suggests contamination or capture bias.
Share of PASS variants (GATK filters)	98.1%	≥ 0.8	Pass	Many filtered variants: degraded sequencing or alignment quality.
Median depth at variants	30×	≥ 15 (clinical threshold)	Pass	Insufficient depth: reduced sensitivity, less reliable genotypes.
Reads mapped to hg38	99.9%	≥ 0.95	Pass	A low rate suggests contamination, a wrong reference or untrimmed adapters.
Duplicate rate (MarkDuplicates)	7.2%	≤ 0.30	Pass	High rate: low library complexity, overestimated useful depth.
Panel ClinVar P/LP sites covered ≥ 15x	99.7%	≥ 0.95	Pass	A poorly covered known pathogenic site cannot be excluded: no variant there does not mean no mutation.

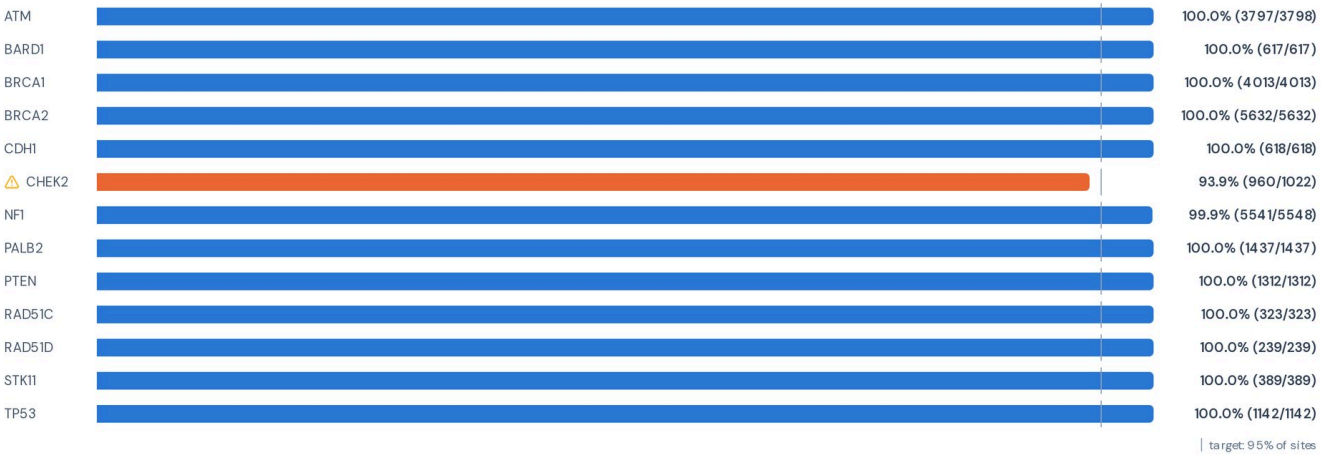
⚠ CHEK2: 960/1022 ClinVar pathogenic sites covered ≥ 15x (93.9%) — an uncovered known mutation cannot be excluded.

Figure 16. Tab *Quality & coverage* — expert quality checks: each metric compared with the range expected for a human germline sample (values explained in chapter 7).

Coverage of known pathogenic sites (≥ 15×)

Table

26,020 / 26,090 ClinVar 2026-09-23 P/LP sites covered (99.7%), median depth 32×. An uncovered site cannot be excluded.



ClinVar pathogenic or likely pathogenic sites (≤ 50 bp) of the panel germline genes; depth with MAPQ ≥ 20 and base quality ≥ 20, minimum over the site's bases.

Figure 17. Coverage of the 26,090 known ClinVar pathogenic sites of the panel at ≥ 15×, per gene. CHEK2 is below the 95 % gene threshold (93.9 %) and is therefore named "not excluded".

BioGPT literature commentary — verified

BioGPT never decides the risk. Every generated sentence is checked against the reference gene–disease associations (ClinGen/NCCN); an unsupported claim is replaced by a deterministic reference sentence.

1/2 generated sentence(s) verified · model `microsoft/biogpt` (base) · knowledge `germlineiq-gene-disease-v2`

CHEK2 ✗ rejected — reference sentence used

Germline pathogenic variants in CHEK2 are associated with an increased risk of breast cancer, prostate cancer and colorectal cancer (ClinGen/NCCN reference associations, germlineiq-gene-disease-v2).

Discarded generated text: "Germline pathogenic variants in CHEK2 are associated with a high risk of breast cancer." — reason: risk magnitude inconsistent with penetrance: 'high risk' but CHEK2 has moderate penetrance

TP53 ✓ BioGPT text verified

Germline pathogenic variants in TP53 are associated with a high risk of breast cancer.

Figure 18. Tab *AI & traceability* — BioGPT literature commentary. BioGPT wrote "CHEK2 … associated with a *high* risk of breast cancer"; the verifier (knowledge v2) rejected it because CHEK2 has moderate penetrance and replaced it by the reference sentence. The TP53 sentence ("high risk") is correct and kept.

BioGPT LoRA fine-tuning

Frozen corpus: 2,001 PubMed abstracts (2005/01/01 → 2026/06/30) collected on 2026-09-26, SHA-256 695a28f1bc62… · 7,077,888 trained parameters out of 353,841,152 (NVIDIA GeForce RTX 5080)

Iteration	Test perplexity	Production prompt: verified	Held-out prompts: unsupported	Promotion
v1 full PubMed corpus (2,001 abstracts) + 3 reference phrasings ×8; lr 2e-4, 4 epochs	15.76 → 8.58	77% → 85%	26% → 51%	rejected
v2 v1 + 8 gene-specific reference phrasings ×12	15.76 → 8.58	77% → 100%	26% → 51%	rejected
v3 v2 + gene-centric abstract filter (987/2,001 kept)	15.03 → 8.56	77% → 77%	26% → 56%	rejected
v4 v3 corpus, light training: lr 5e-5, 2 epochs	15.03 → 9.32	77% → 77%	26% → 23%	rejected

Decision: no adapter promoted automatically (criterion: no factual metric may degrade). In service: base BioGPT + gene–disease verification + deterministic fallback sentence.

13 generations (production prompt) and 39 (held-out prompts): 1 generation = 7.7 % / 2.6 %; differences between iterations are within noise.

Figure 19. Literature-commentary fine-tuning history (four LoRA iterations on frozen PubMed abstracts). Perplexity improved but factual accuracy on held-out prompts did not, so no commentary adapter was promoted: the gate protects production.

Reproducibility

VCF fingerprint (SHA-256)
`cf3c1ddae5066de85a9adb898af70cdd2c59cab167e38e1d8843fa1c3d26e1df`

Panel version
`d8308ee353a5`

Annotation
`clinvar 2026-09-23`

Decision method
`germlineiq-rules-v1.1`

Statistics
`germlineiq-stats-v1`

Software
`3.0.0`

Clinical thresholds: min_qual = 30 · min_depth = 15 · het_vaf_min = 0.25 · het_vaf_max = 0.75 · hom_vaf_min = 0.85

Figure 20. Reproducibility block: input SHA-256, panel, ClinVar release, QC thresholds, decision method and software version.

System metrics

EXECUTION TIME
7.3 s

HARDWARE
Local server (6 cores) — NVIDIA GeForce RTX 5080 (16 GB)

PIPELINE ENGINE
NVIDIA Clara Parabricks + GATK (GPU)

PATIENT ID
NA10842

REPORT ID
REP-NA10842-20260927-cf3c1dda

GENERATED AT
2026-09-27T13:32:16Z

COMPLETED STEPS

Data preparation | Variant calling | ClinVar annotation | Panel analysis | Interpretation | Report

Figure 21. System metrics: engine (Parabricks + GATK on GPU), orchestration plan, router and hardware.

Cohort page

GermlineIQ
Hereditary cancer germline analysis

Analyze | Cohort | Benchmarks | Architecture

API online | Parabricks GPU · LangGraph (mistral)

Analysis history

Cohort

Every analysis run on this server. The demonstration cohort contains real open-access genomes from the 1000 Genomes Project (30× reads over the panel genes) and the GIAB NA12878 reference exome.

Refresh

Patients analysed	Risk HIGH	Risk MODERATE	Risk INDETERMINATE	Risk LOW
15	6	4	1	4
35 runs shown	latest run per patient	latest run per patient	latest run per patient	latest run per patient

All | HIGH | MODERATE | INDETERMINATE | LOW | Failed | ☑ Latest run per patient ☐ Show test & benchmark runs

Patient	Mode	Risk	Genes with P/LP variants	Panel variants	Duration	Started
REPORT_DEMO	VCF	INDETERMINATE	none	1	14.3 s	27 Sept, 19:41
DEMO_HIGH	VCF	HIGH	BRCA1	3	8.6 s	27 Sept, 16:11
NA10842_DEMO	FASTQ	MODERATE	CHEK2	1273	7.8 s	27 Sept, 16:02
HG02620_DEMO	FASTQ	HIGH	PALB2	1599	7.5 s	27 Sept, 15:59
NA19130	FASTQ	HIGH	BRCA2	1853	7.4 s	27 Sept, 15:36
HG01112	FASTQ	LOW	none	1395	1 min 35 s	27 Sept, 15:34
HG00096	FASTQ	LOW	none	1634	4.5 s	27 Sept, 15:33
HG00596	FASTQ	MODERATE	ATM	1332	1 min 39 s	27 Sept, 15:32
NA10842	FASTQ	MODERATE	CHEK2	1273	7.5 s	27 Sept, 15:32
HG02620	FASTQ	HIGH	PALB2	1599	1 min 39 s	27 Sept, 15:30
HG00611	FASTQ	HIGH	BRCA2	1553	1 min 39 s	27 Sept, 15:28
NA12878_GPU	FASTQ	LOW	none	192	7.1 s	26 Sept, 22:55
NA10842_GPU	FASTQ	MODERATE	CHEK2	1273	1 min 50 s	26 Sept, 21:26
DEMO_BRCA1	VCF	HIGH	BRCA1	3	9.4 s	26 Sept, 21:10
NA12878	FASTQ	LOW	none	192	58 min 49 s	26 Sept, 17:52

GermlineIQ · GATK · ClinVar · deterministic risk rules · verified BioGPT commentary · for clinical research use only

Figure 22. Cohort page: latest analysis per patient with risk level, genes carrying P/LP variants, number of panel variants and duration. Tiles count patients per risk level; test and benchmark runs are hidden by default.

API

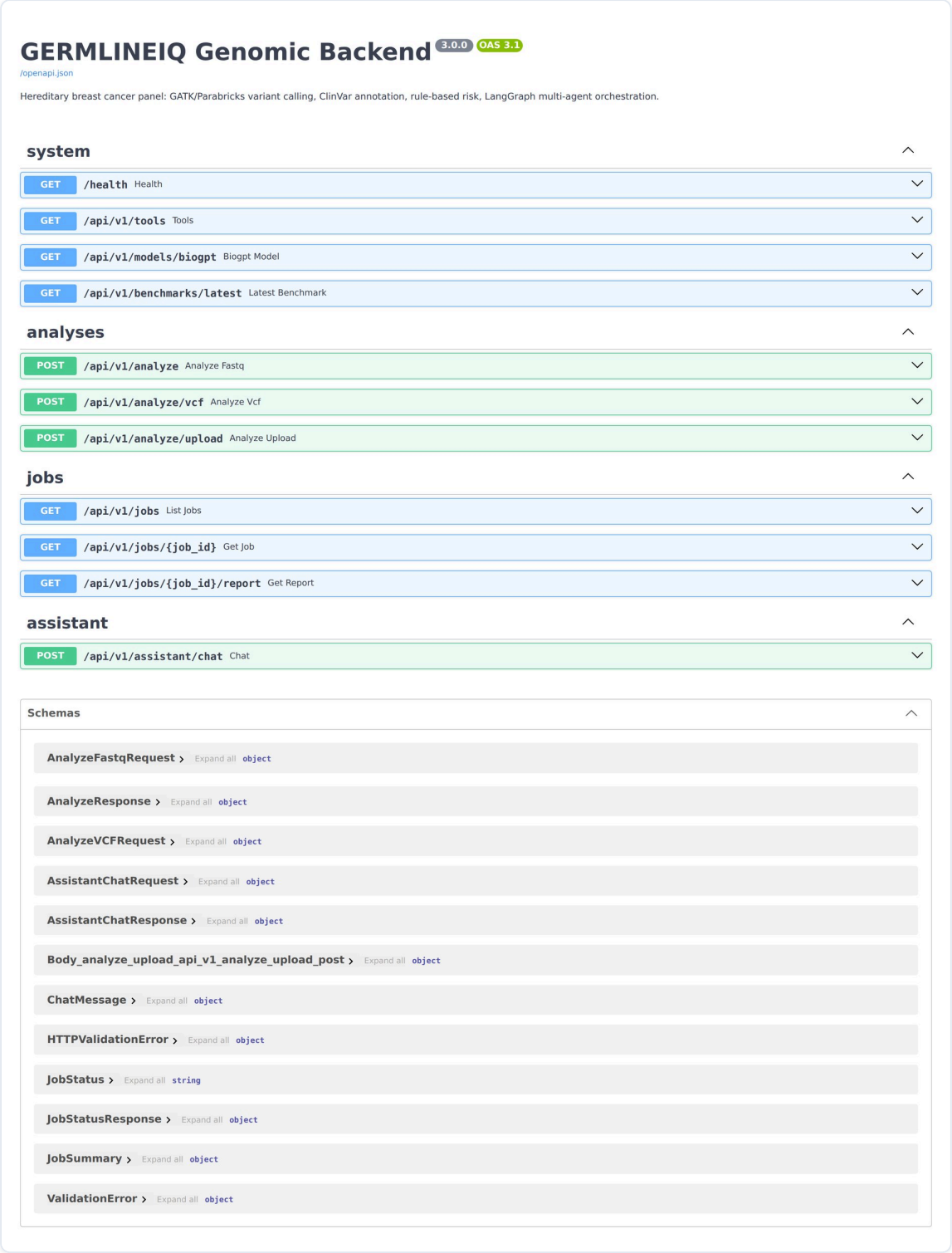


Figure 23. Interactive OpenAPI documentation of the FastAPI backend (`http://localhost:8000/docs`): analysis endpoints (FASTQ, VCF, upload), jobs and reports, model and benchmark information, assistant.

5. Technical description of every tool

Agents are registered in a tool registry (`src/orchestration/registry.py`) that declares, for each tool, the context keys it *requires* and *produces*, its GPU phase, and which inputs and configuration variables form its cache key. The planner derives the plan from these declarations; the table-driven design is what lets a VCF input skip alignment automatically.

`data_manager` — DataManagerAgent	
Purpose	Validate and register the FASTQ pair of a patient.
Inputs → outputs	`patient_id, fastq_r1, fastq_r2` → `fastq_r1_uri, fastq_r2_uri`
Implementation	Path validation (existing file, under `LOCAL_DATA_ROOT`, FASTQ extension, R1 ≠ R2, S3 URIs refused) and storage in `patients/<ID>/input/`.
Key parameters	Patient ID pattern; data root.
Artifact	`patients/<ID>/input/`

`genomic_pipeline` — VariantCallingAgent (tool version 4)	
Purpose	FASTQ → filtered germline VCF on the panel, alignment QC and coverage of known pathogenic sites.
Inputs → outputs	`fastq_r1_uri, fastq_r2_uri` → `vcf_uri, bam_uri, pipeline_backend, alignment_qc`
Implementation	**GPU (Parabricks 4.6.0-1):** `pbrun fq2bam` (GPU BWA-MEM, coordinate sort, duplicate marking, BQSR table with known indels + Mills) then `pbrun haplotypecaller` restricted to the panel BED, applying the recalibration on the fly. **CPU (GATK 4.2.6.1):** `bwa mem -K 100000000 -Y` → `samtools sort` → MarkDuplicates → BaseRecalibrator/ApplyBQSR → HaplotypeCaller `-L panel`. **Common post-processing:** LeftAlignAndTrimVariants with multi-allelic splitting, then GATK hard filters per variant type.
Key parameters	Panel ±100 bp; SNP filters QD<2, QUAL<30, SOR>3, FS>60, MQ<40, MQRankSum<−12.5, ReadPosRankSum<−8; indel filters QD<2, QUAL<30, FS>200, ReadPosRankSum<−20. Engine selection: Parabricks if VRAM ≥ 16 GB, automatic fallback to GATK4 if Parabricks fails. Containers run as the service user.
Artifact	`aligned.bam, variants.vcf.gz, duplicate_metrics.txt, alignment_qc.json`
Alignment QC	`samtools flagstat` (mapping, pairing), Picard-format duplicate metrics, `samtools depth -a -Q20 -q20` at every ClinVar P/LP site (≤ 50 bp) of the 13 germline genes; step checkpoints for crash recovery.

`variant_annotation` — VariantAnnotationAgent	
Purpose	Annotate panel variants with ClinVar.
Inputs → outputs	`vcf_uri` → `annotated_variants_path, annotation, input_sha256`
Implementation	Local ClinVar GRCh38 VCF indexed once per panel version; exact allele match after normalisation; review status converted to stars. The input VCF SHA-256 is recorded for traceability.
Key parameters	ClinVar release 2026-09-23; panel d8308ee353a5.
Artifact	`annotated_variants.json`

`vcf_analysis` — VCFAnalysisAgent	
Purpose	Clinical QC, classification of every variant and descriptive statistics.
Inputs → outputs	`annotated_variants_path` → `panel_analysis, vcf_metrics, vcf_statistics`
Implementation	Categories: pathogenic confirmed · pathogenic to confirm · loss of function to confirm · conflicting · VUS · benign · likely benign · not in ClinVar. Statistics `germlineiq-stats-v1`: type-7 quantiles, fixed-bin histograms (comparable across patients), Ti/Tv, het/hom, VAF, per-gene tables and expert checks against expected ranges.
Key parameters	Clinical QC: QUAL ≥ 30, DP ≥ 15, heterozygous VAF 0.25–0.75, homozygous VAF ≥ 0.85, FILTER = PASS; otherwise a P/LP variant is "to confirm".
Artifact	`panel_analysis.json, vcf_statistics.json`

`prediction` — PredictionAgent	
Purpose	Risk decision and verified texts.
Inputs → outputs	`panel_analysis (+ vcf_statistics, alignment_qc)` → `risk_assessment, prediction_results`
Implementation	**Rules germlineiq-rules-v1.1** (risk-rules table below). **BioGPT literature commentary** verified against curated ClinGen/NCCN gene–disease associations and gene penetrance (knowledge v2). **BioGPT statistics interpreter** (LoRA adapter v1): facts → generation → sentence verification → final answer with reference fallback (chapter 6).
Key parameters	BioGPT greedy decoding (deterministic); models loaded then unloaded to free VRAM.
Artifact	`risk decision inside the report`

`report` — ReportGeneratorAgent	
Purpose	Assemble and archive the clinical report.
Inputs → outputs	`panel_analysis, prediction_results` → `clinical_report, report_uri`
Implementation	Pydantic schema (`ClinicalReport`): genomic findings, clinical prediction, statistics, reproducibility, system metrics; legal disclaimer; status AWAITING_MEDICAL_VALIDATION.
Key parameters	—
Artifact	`REP-<ID>-<date>-<sha>.json`

Table 5. Deterministic risk rules. The rules never depend on a language model.

Level	Rule (germlineiq-rules-v1.1)
HIGH	≥ 1 confirmed P/LP variant (ClinVar) in a high-penetrance gene (BRCA1, BRCA2, PALB2, TP53, PTEN, CDH1, STK11)
MODERATE	≥ 1 confirmed P/LP variant in a moderate-penetrance gene (CHEK2, ATM, BARD1, RAD51C, RAD51D, NF1)
INDETERMINATE	No confirmed variant but a P/LP variant to confirm (QC failure, unclassified loss of function) — or, in FASTQ mode, < 90 % of the panel's known pathogenic sites covered ≥ 15×
LOW	No P/LP variant and nothing to confirm; genes with < 95 % of their pathogenic sites covered are named "not excluded"

Orchestration, interfaces and verification tools

Component	Technology	Role and behaviour
Orchestrator	LangGraph	State graph *plan ⇄ execute*: the planner recomputes the tools still needed from the current context after each step; context merging is deterministic.
Router	Rules or Mistral v0.3 (Ollama)	Chooses the next tool among the *ready* tools only; the benchmark shows both routers produce the same plan and the same report.
Result cache	SHA-256 key	Tool + version + input file fingerprints + configuration variables + panel; stale entries detected when files disappear.
GPU manager	nvidia-smi + lock	Real VRAM detection, mutual exclusion between Parabricks and BioGPT, model unloading after use.
API	FastAPI 0.141 · Pydantic v2	Analysis (FASTQ, VCF, upload), jobs and reports, BioGPT model information, benchmark results, assistant; persisted job queue (sequential by design to protect VRAM).
Assistant	Mistral v0.3 via Ollama	Intent parsing (start FASTQ / start VCF / job status), missing-field prompts; same validation as the API.
MCP server	stdio	Exposes the same tools to MCP-compatible clients.
Interface	Next.js 14 · Tailwind	Analyze, Cohort, Benchmarks, Architecture; report dashboards with accessible charts (table view for every chart).
Independent verification	bcftools · samtools · awk	`scripts/expert_check.py` recomputes 20 report metrics without importing backend code.
Benchmark	Python client of the API	`benchmarks/multiagent.py`: clinical, analytical, reproducibility, orchestration, robustness, latency suites.
Operations	systemd user service	Ollama + backend + frontend start automatically; Docker image on PyTorch 2.8 / CUDA 12.8 (Blackwell-ready).

Table 6. Software and data versions of the validation run.

Software	Version	Software	Version
GATK (container)	4.2.6.1	Python	3.11.16
NVIDIA Parabricks	4.6.0-1	PyTorch	2.11.0 (CUDA 12.8)
BWA	0.7.19	Transformers / PEFT	5.17.0 / 0.21.0
samtools / bcftools	1.24 / 1.24	LangGraph / FastAPI	1.2.12 / 0.141.1
BioGPT	microsoft/biogpt + LoRA stats adapter v1	Ollama / Mistral	0.34.4 / mistral:v0.3
ClinVar (GRCh38)	2026-09-23	Node.js / Next.js	20.20 / 14.2
GIAB truth set	HG001 GRCh38 v4.2.1	NVIDIA driver	591.74 (RTX 5080, sm_120)

6. Validation: benchmarks and metrics

Metrics used

Table 7. Metrics and their clinical meaning.

Metric	Definition	Why it matters
Sensitivity (recall)	TP / (TP + FN)	Proportion of true pathogenic variants / carriers found; a missed carrier is the most harmful error.
Specificity	TN / (TN + FP)	Proportion of non-carriers correctly reported without a P/LP finding.
Precision (PPV)	TP / (TP + FP)	Proportion of reported calls that are true; drives unnecessary confirmations.
F1 score	2·P·R / (P + R)	Single summary of precision and recall for variant calling.
Wilson 95 % CI	Score interval for a binomial proportion	Honest uncertainty on small cohorts (n = 7 gives wide intervals by construction).
Ti/Tv	Transitions / transversions (PASS SNVs)	≈ 2.0–2.1 genome-wide, ≈ 2.8–3.0 in coding exons; low values indicate false-positive calls.
Het/hom ratio	Heterozygous / homozygous-alt calls	≈ 1.4–2.2 depending on ancestry; high values may indicate contamination.
Heterozygous VAF	Alt reads / total reads at het calls	≈ 0.5 for germline heterozygotes; a shift or widening suggests contamination or somatic mosaicism.
Site coverage	Known P/LP sites with depth ≥ 15×	An uncovered known pathogenic site cannot be excluded; gates a LOW conclusion.
Sentence precision	Verified / generated sentences	Factual reliability of BioGPT text measured by the deterministic verifier.
Topic coverage	Required topics answered by a verified model sentence	Completeness of the model's interpretation.

Clinical accuracy on a real open-access cohort

Seven genomes of the 1000 Genomes Project (30× Illumina, New York Genome Center) were selected by an independent scan of the 1000 Genomes germline call set against ClinVar P/LP variants in the 13 panel genes: five carriers (two BRCA2, one PALB2, one CHEK2, one ATM) and two controls. Real reads over the panel genes were streamed from the public CRAM files and processed end to end through the API (FASTQ → report) on the GPU pipeline.

Table 8. Per-sample results of the clinical suite (final benchmark, Parabricks GPU). Short times = variant calling reused from the result cache.

Sample	Ground truth	Expected variant	Detected P/LP	Risk (exp → got)	Correct	P/LP sites ≥15×	Time
NA19130	BRCA2 pathogenic insertion (ClinVar expert panel)	`chr13:32339596:C:CTA`	`chr13:32339596:C:CTA`	HIGH → **HIGH**	✔ PASS	99.98 %	111 s
HG00611	BRCA2 pathogenic duplication (ClinVar expert panel)	`chr13:32357820:G:GA`	`chr13:32357820:G:GA`	HIGH → **HIGH**	✔ PASS	100.00 %	102 s

Sample	Ground truth	Expected variant	Detected P/LP	Risk (exp → got)	Correct	P/LP sites ≥15×	Time
HG02620	PALB2 pathogenic deletion	`chr16:23636312:AAC:A`	`chr16:23636312:AAC:A`	HIGH → **HIGH**	✔ PASS	99.97 %	99 s
NA10842	CHEK2 c.1100delC founder variant	`chr22:28695868:AG:A`	`chr17:7673802:C:T`, `chr22:28695868:AG:A`	MODERATE → **MODERATE**	✔ PASS	99.73 %	9 s
HG00596	ATM pathogenic variant	`chr11:108271407:G:T`	`chr11:108271407:G:T`	MODERATE → **MODERATE**	✔ PASS	100.00 %	99 s
HG00096	No ClinVar P/LP variant in the panel (control)	–	–	LOW → **LOW**	✔ PASS	99.99 %	6 s
HG01112	No ClinVar P/LP variant in the panel (control)	–	–	LOW → **LOW**	✔ PASS	99.98 %	96 s

Table 9. Summary metrics of the clinical suite.

Metric	Value	95 % Wilson CI
Carrier sensitivity	100 % (5/5)	56.5–100 %
Control specificity	100 % (2/2)	34.2–100 %
Risk-level accuracy	100 % (7/7)	64.6–100 %
Variant recall (expected P/LP alleles)	100 %	—
Unexpected confirmed P/LP	0	—
Unexpected calls flagged “to confirm”	1	TP53 p.Arg273His at VAF 0.14 in NA10842
End-to-end time (FASTQ → report)	median 99 s · max 111 s	≈150k read pairs per sample

The TP53 call in NA10842. HaplotypeCaller found TP53 c.818G>A (p.Arg273His, rs28934576, ClinVar Pathogenic, expert panel) in 5 of 36 reads (VAF 0.139), filtered by GATK (QD < 2). It is absent from the 1000 Genomes germline call set. A low-VAF hotspot mutation in lymphoblastoid-cell-line DNA is typical of a culture-acquired or clonal-haematopoiesis event rather than a germline variant. GermlineIQ did exactly what a clinical laboratory would require: it did not count the call towards the risk, and reported it “to confirm” with its reasons (FILTER:SNP_QD2, LOW_VAF_MOSAIC_OR_CHIP).

Clinical accuracy — real 1000 Genomes cohort

1000 Genomes Project, 30x high-coverage (NYGC), open access — IGSR. Real Illumina reads over the 13 panel genes, full FASTQ → report pipeline. Ground truth: independent bcftools scan of the 1000 Genomes germline calls against ClinVar P/LP. n = 7: confidence intervals are wide by design. The TP53 p.Arg273His call in NA10842 (c.818G>A, rs28934576; 5/36 reads, VAF 0.14) is absent from the germline call set: a low-VAF hotspot in lymphoblastoid-cell-line DNA is typical of a culture-acquired or clonal-haematopoiesis mutation, and GermlineIQ correctly reports it "to confirm" rather than as a germline finding.

Sample	Ground truth	Reported by GermlineIQ	Risk (expected → reported)	P/LP sites ≥15×	End-to-end
NA19130 BRCA2 pathogenic insertion (ClinVar expert panel)	chr13:32339596:C:CTA	**BRCA2** C>CTA · heterozygous · DP 36 · VAF 0.42	HIGH → **HIGH**	100.0%	111 s
HG00611 BRCA2 pathogenic duplication (ClinVar expert panel)	chr13:32357820:G:GA	**BRCA2** G>GA · heterozygous · DP 46 · VAF 0.41	HIGH → **HIGH**	100.0%	102 s
HG02620 PALB2 pathogenic deletion	chr16:23636312:AAC:A	**PALB2** AAC>A · heterozygous · DP 19 · VAF 0.68	HIGH → **HIGH**	100.0%	99 s
NA10842 CHEK2 c.1100delC founder variant	chr22:28695868:AG:A	**CHEK2** AG>A · heterozygous · DP 36 · VAF 0.39 **TP53** C>T · heterozygous · DP 36 · VAF 0.14 to confirm	MODERATE → **MODERATE**	99.7%	9 s
HG00596 ATM pathogenic variant	chr11:108271407:G:T	**ATM** G>T · heterozygous · DP 43 · VAF 0.51	MODERATE → **MODERATE**	100.0%	99 s
HG00096 No ClinVar P/LP variant in the panel (control)	–	no P/LP variant	LOW → **LOW**	100.0%	6 s
HG01112 No ClinVar P/LP variant in the panel (control)	–	no P/LP variant	LOW → **LOW**	100.0%	96 s

Figure 24. Benchmarks page — clinical accuracy section as displayed to users.

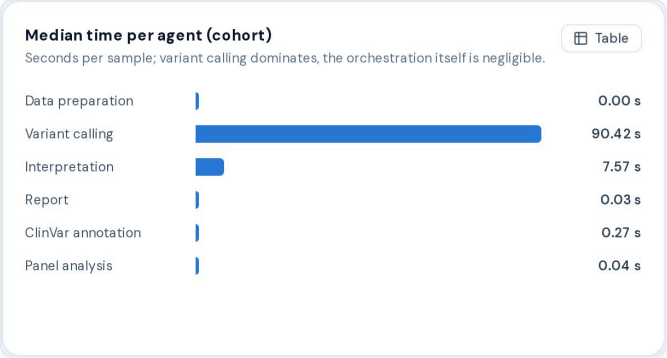


Figure 25. Median time per agent over the cohort.

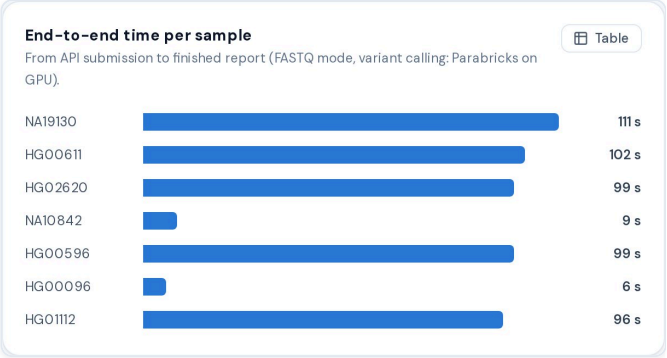


Figure 26. End-to-end time per sample (FASTQ mode, Parabricks GPU).

Reading the timing charts. Variant calling dominates: median 90 s over the five samples that ran it (fq2bam ≈ 45 s, HaplotypeCaller ≈ 17 s, GATK post-processing ≈ 30–45 s). Clinical interpretation takes a median 7.6 s (loading BioGPT twice — commentary and statistics — then releasing VRAM). Annotation (0.27 s), analysis (0.04 s), data preparation and report assembly are negligible: the orchestration itself costs a fraction of a second. On inputs this small (≈150k read pairs) the GPU is not faster than GATK4 on CPU (median 93 s in the CPU benchmark), because loading the 5 GB BWA index into VRAM is a fixed cost; the GPU advantage grows with exome and genome sizes.

Analytical validity — GIAB HG001 (NA12878)

The NA12878 exome (SRR1611178, 2 × 4 GB FASTQ, 90,115,545 reads) was processed through the pipeline and compared with the Genome in a Bottle v4.2.1 truth set on 131,040 bp: panel genes ± 100 bp ∩ GIAB high-confidence ∩ depth ≥ 15x.

Table 10. Concordance with GIAB HG001 v4.2.1 (small variants).

Calls	TP	FP	FN	Precision	Recall (95 % CI)	F1
PASS (GATK hard filters)	79	2	10	97.5 %	88.8 % (80.5 %–93.8 %)	0.929
All HaplotypeCaller calls	89	6	0	93.7 %	100 % (95.9 %–100 %)	0.967

Every truth variant missed at the PASS level was detected by HaplotypeCaller (100 % recall on all calls) and removed by a hard filter — mostly the strand-odds-ratio filter (SOR > 3), a known artefact of exome capture. The fil-

ters were deliberately kept at GATK defaults: pathogenic variants among filtered calls are never discarded but reported “to confirm”, which preserves clinical sensitivity while keeping precision high.

Analytical validity — GIAB HG001 (NA12878)

NA12878 / GIAB HG001 v4.2.1 (exome SRR1611178). Region: panel genes ± 100 bp ∩ GIAB high-confidence ∩ depth ≥ 15x (131,040 bp).

Calls	TP	FP	FN	Precision	Recall	F1
PASS (GATK hard filters)	79	2	10	97.5%	88.8%	0.929
All HaplotypeCaller calls	89	6	0	93.7%	100.0%	0.967

Filtered pathogenic variants are never discarded: they are reported “to confirm”, which keeps clinical sensitivity even when a hard filter rejects a call. Every call missed at the PASS level was detected by HaplotypeCaller (100% recall on all calls) and removed by GATK hard filters, mostly the SOR strand-bias filter — a known exome-capture artefact.

Figure 27. Benchmarks page — analytical validity section.

Verified AI — BioGPT fine-tuned on VCF statistics

Workflow. (1) *Training:* 4,000 statistics files sampled over normal and abnormal ranges (low Ti/Tv, contamination-like het/hom and VAF, low depth, poor mapping, high duplication, insufficient site coverage), with statuses and risk levels computed by the pipeline's own thresholds and rules; targets are reference interpretations with paraphrases; LoRA (rank 16, 7,077,888 trainable of 353,841,152 parameters), loss on interpretation tokens only, 256 s on the NVIDIA GeForce RTX 5080. (2) *Integration:* the adapter is promoted only if it passes the gate on both test sets. (3) *Prediction* on each new `vcf_statistics.json`. (4) *Deterministic verification* of every sentence. (5) *Final answer*: verified model sentence per topic, else reference sentence.

Table 11. Evaluation of the statistics interpreter through the deterministic verifier. Gate: sentence precision ≥ 95 %, topic coverage ≥ 90 %, no wrong risk level, every final answer verified — passed on both sets.

Test set	Model	n	Verified sentences	Topics covered	Numbers correct	Status words correct	Wrong risk levels
Synthetic statistics (held out)	Base BioGPT (zero-shot)	40	0.0 %	0.0 %	—	—	2
Synthetic statistics (held out)	**Fine-tuned (LoRA v1)**	300	**98.6 %**	98.6 %	100.0 %	97.8 %	0
Real analyses (never trained on)	Base BioGPT (zero-shot)	15	6.7 %	0.6 %	—	—	1
Real analyses (never trained on)	**Fine-tuned (LoRA v1)**	15	**100.0 %**	100.0 %	100.0 %	100.0 %	0

The base model cannot perform the task (it generates generic biomedical text; 0 % verified on synthetic files). After fine-tuning, numbers are always correct; the residual errors (1.4 % of sentences on synthetic data) are wrong status words — e.g. “median depth 10x, adequate” while depth is below the 15× threshold — and they are caught by the verifier and replaced. **Honest limitation:** the training targets are written by a rule-based expert interpreter, so the model learns to *verbalise* statistics fluently and reliably; it does not discover new knowledge.

BioGPT fine-tuned on VCF statistics — verified interpretation

Every generated sentence is scored by the deterministic verifier (numbers, expected ranges, risk level, genes). Training: 4000 statistics files, 256 s on NVIDIA GeForce RTX 5080. Promotion gate: sentence precision ≥ 95%, topic coverage ≥ 90%, no wrong risk level, every final answer verified.

Test set	Model	n	Verified sentences	Topics covered	Numbers correct	Status words correct	Wrong risk levels
Synthetic statistics files (normal and abnormal QC)	Base BioGPT (zero-shot)	40	0.0%	0.0%	–	–	2
Synthetic statistics files (normal and abnormal QC)	Fine-tuned (LoRA)	300	98.6%	98.6%	100.0%	97.8%	0
Real analyses of this server (held out)	Base BioGPT (zero-shot)	15	6.7%	0.6%	–	–	1
Real analyses of this server (held out)	Fine-tuned (LoRA)	15	100.0%	100.0%	100.0%	100.0%	0

Promotion gate passed — adapter v1 in service Rejected sentences are replaced by the deterministic reference text, so every final interpretation is verified.

Figure 28. Benchmarks page — evaluation of the statistics interpreter.

Table 12. Literature-commentary LoRA iterations (PubMed corpus). Lower perplexity did not translate into better factual accuracy on held-out prompts, so no commentary adapter was promoted; production uses base BioGPT with sentence verification.

Iteration	Description	Test perplexity (base → tuned)	Verified, held-out prompts (base → tuned)
v1	full PubMed corpus (2,001 abstracts) + 3 reference phrasings ×8; lr 2e-4, 4 epochs	15.8 → 8.6	69 % → 46 %
v2	v1 + 8 gene-specific reference phrasings ×12	15.8 → 8.6	69 % → 46 %
v3	v2 + gene-centric abstract filter (987/2,001 kept)	15.0 → 8.6	69 % → 44 %
v4	v3 corpus, light training: lr 5e-5, 2 epochs	15.0 → 9.3	69 % → 72 %

Determinism, orchestration, robustness and latency

Table 13. Reproducibility: SHA-256 of the clinical content (genomic findings, prediction — including the BioGPT interpretation — statistics, reproducibility block; timestamps and timings excluded).

Case	Runs	Identical reports	Identical statistics	Report SHA-256 (prefix)
VCF — synthetic BRCA1/BRCA2	3	✔ PASS	✔ PASS	`e5ba893630d86238…`
FASTQ — NA19130 (cached variant calling)	3	✔ PASS	✔ PASS	`e84668e880e950cd…`

Table 14. Deterministic versus Mistral router on the same VCF (three runs each): same plan and byte-identical clinical report.

Router	Plan	Total (median)	Orchestration overhead (median)	Same report
deterministic	variant_annotation → vcf_analysis → prediction → report	7.80 s	0.198 s	✔ PASS
mistral	variant_annotation → vcf_analysis → prediction → report	6.77 s	0.163 s	✔ PASS

Table 15. Robustness (fault injection).

Fault injected	Expected	Observed	
invalid patient id	422	422	✔ PASS
path outside the data root	422	422	✔ PASS
missing file	422	422	✔ PASS
identical R1 and R2	422	422	✔ PASS
S3 URI rejected	422	422	✔ PASS
unknown job id	404	404	✔ PASS
empty body	422	422	✔ PASS
malformed VCF job	failed	failed	✔ PASS
service healthy after faults	200	200	✔ PASS

Table 16. API latency (50 sequential requests after warm-up).

Endpoint	p50	p95	p99
GET /health	0.95 ms	1.10 ms	1.28 ms
GET /api/v1/tools	0.83 ms	0.91 ms	0.94 ms
GET /api/v1/jobs?limit=50	0.98 ms	1.10 ms	1.13 ms
GET /api/v1/models/biogpt	1.27 ms	1.48 ms	1.56 ms

GPU versus CPU concordance and independent verification

Table 17. Engine concordance and independent recomputation of every report metric with bcftools/samtools.

Comparison	Result
NA10842 — final filtered VCF, Parabricks (GPU) vs GATK4 (CPU)	1,357 / 1,357 calls shared, 0 discordant; 1,333 / 1,333 PASS; identical genotypes and QUAL (CHEK2 c.1100delC QUAL 450.6 on both)
HG00096 — raw HaplotypeCaller calls, GPU vs CPU	1,704 shared calls with identical genotypes, 0 GPU-only, 2 CPU-only
Independent verification (`expert_check.py`)	NA19130 20/20; HG00611 20/20; HG02620 20/20; NA10842 20/20; HG00596 20/20; HG00096 20/20; HG01112 20/20; NA12878 20/20

Table 18. Independent verification of the NA10842 report: every value recomputed from the VCF, BAM and ClinVar with reference tools, without backend code.

Check (NA10842)	Report	Independent	Status	Method
All strict-gene variants reported	0	0	✔ PASS	1271 inside the genes (no padding)
All reported variants within gene ± padding	0	0	✔ PASS	2 in the 100 bp padding (gene assigned by ClinVar)
Germline panel variants	1273	1273	✔ PASS	bcftools view -R | norm -m -any | GT=alt
PASS variants	1249	1249	✔ PASS	
PASS SNV transitions	629	629	✔ PASS	bcftools stats (TSTV)
PASS SNV transversions	308	308	✔ PASS	bcftools stats (TSTV)
Ti/Tv	2.042	2.042	✔ PASS	
Het/hom ratio	2.12	2.12	✔ PASS	
Median depth (FORMAT/DP)	30.0	30	✔ PASS	
Median heterozygous VAF (AD)	0.5	0.5	✔ PASS	
Total reads	286917	286917	✔ PASS	samtools flagstat
Mapping rate	0.9991	0.9991	✔ PASS	
Duplicate rate (Picard)	0.072	0.072	✔ PASS	awk on duplicate_metrics.txt
ClinVar P/LP sites assessed	26090	26090	✔ PASS	bcftools + awk (primary CLNSIG, ALT ≠ ".")
P/LP sites covered ≥ 15x	26020	26020	✔ PASS	samtools depth -a -Q20 -q20
Confirmed P/LP (report = statistics)	1	1	✔ PASS	
To confirm (report = statistics)	1	1	✔ PASS	
VUS (report = statistics)	0	0	✔ PASS	
Panel variants (report = statistics)	1273	1273	✔ PASS	
Risk level consistent with P/LP	MODERATE	MODERATE	✔ PASS	

During the preparation of this report the verification script itself was improved: when variant calling is reused from the result cache (same FASTQ files analysed under another patient ID), it now verifies the BAM and VCF the report actually used, and says so. Before this change it compared the report with an older CPU alignment left in the folder, which produced a 6-site discrepancy on the coverage count (26,020 vs 26,014) — a checker issue, not a report error.

7. Reading the charts: genomic values explained

This chapter reads each chart of the *VCF statistics* and *Quality & coverage* tabs with the values measured on two real samples: **NA10842** (1000 Genomes, 30× genome restricted to the panel genes) and **NA12878** (GIAB exome). Bins are fixed across patients so that charts are directly comparable.

NA10842 — overview

Table 19. Key values of NA10842 (from vcf_statistics.json).

Quantity	Value	Quantity	Value
Panel variants (alleles)	1,273	PASS calls	1,249 (98.1 %)
SNVs / insertions / deletions	960 / 148 / 165	Heterozygous / homozygous	865 / 408
Ti/Tv (PASS SNVs)	2.042 (629 / 308)	Het/hom ratio	2.12
ClinVar-annotated	271 (21.3 %)	Pass clinical QC	1,159 (91.0 %)
Reads / mapped / properly paired	286,917 / 99.91 % / 99.71 %	Duplicate rate	7.2 %

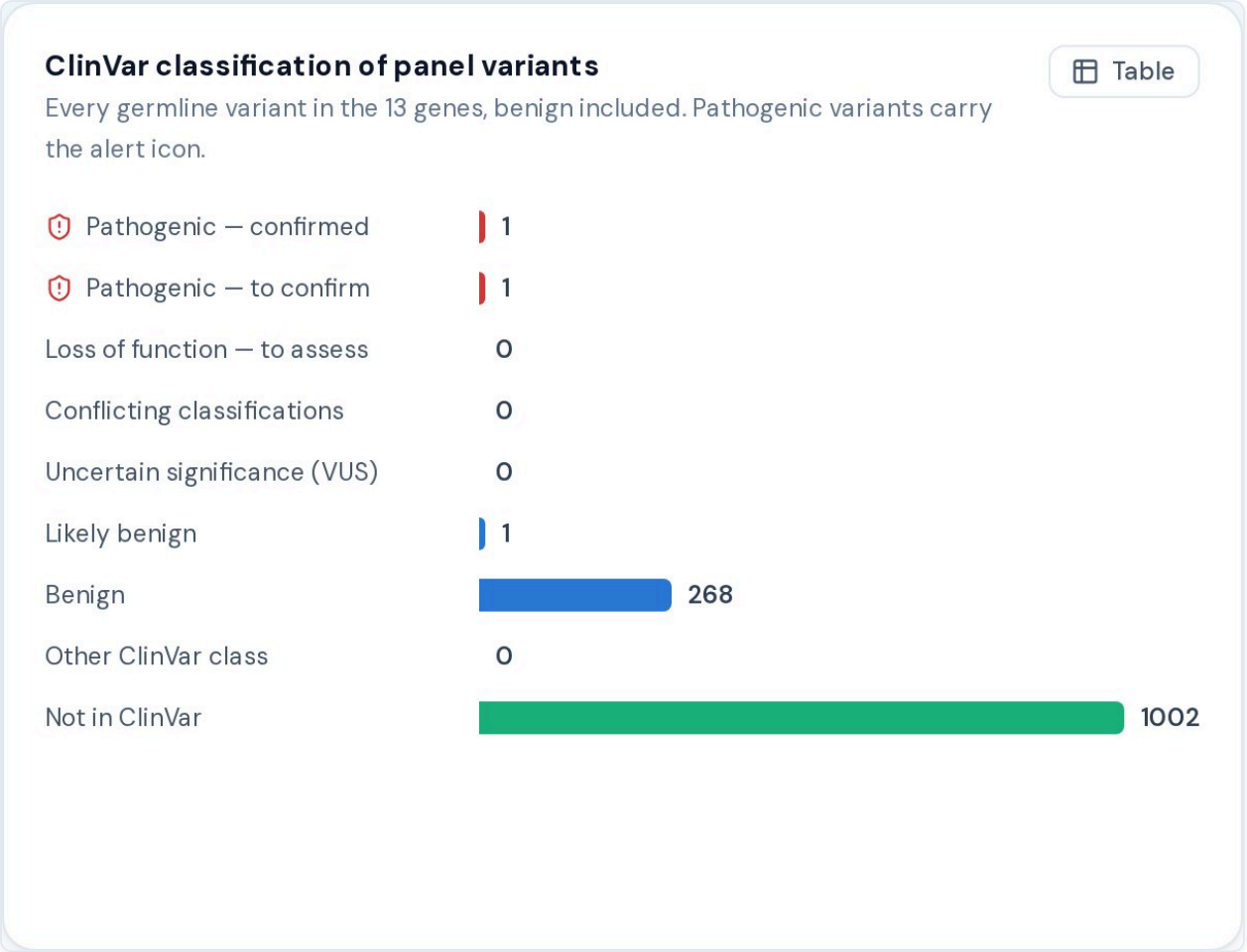


Figure 29. ClinVar classification of the panel variants of NA10842.

ClinVar classification. Of 1,273 variants, 1,002 (79 %) are absent from ClinVar — mostly intronic and common population variants that no laboratory submits —, 268 benign and 1 likely benign, 0 VUS, 0 with conflicting interpretations. Only two are pathogenic: CHEK2 c.1100delC (confirmed) and TP53 p.Arg273His (to confirm). This is the

expected profile of a healthy-looking genome: clinically relevant variants are rare needles in a large benign background, which is why the classification must be exact-allele and versioned.

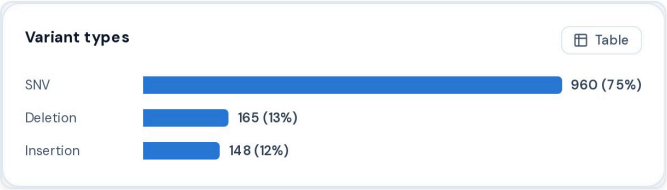


Figure 30. Variant types.

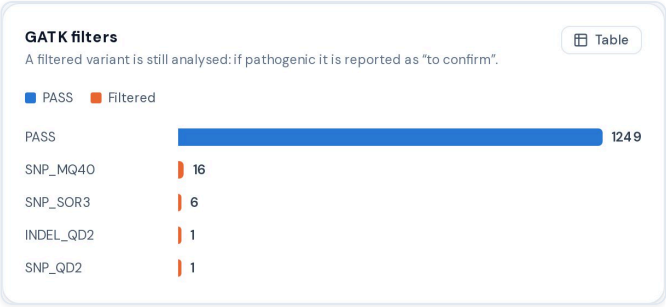


Figure 31. GATK filters.

Variant types and filters. SNVs represent 75 % of calls and indels 25 %, a typical ratio for gene bodies that include introns and low-complexity repeats. 1,249 calls pass all hard filters (98.1 %, above the 80 % check); the filtered calls are INDEL QD2 (1), SNP MQ40 (16), SNP QD2 (1), SNP SOR3 (6). MQ < 40 flags reads that map ambiguously (segmental duplications and pseudogenes), QD < 2 low confidence relative to depth — the filter that caught the TP53 call.

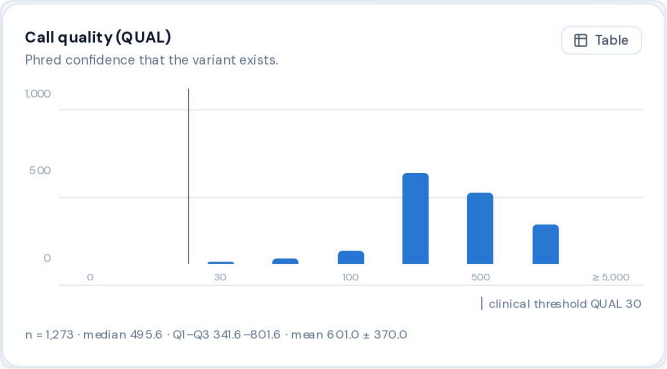


Figure 32. Call quality (QUAL).

Figure 33. Genotype quality (GQ).

Table 20. NA10842 distributions (type-7 quantiles).

Distribution	n	Min	Q1	Median	Q3	Max	Mean ± SD
QUAL (all calls)	1,273	31.6	341.6	**495.6**	801.6	2,208.1	601.0 ± 370.0
QUAL (PASS only)	1,249	31.6	348.6	**503.6**	822.1	2,208.1	607.5 ± 370.0
Read depth DP	1,273	3.0	24.0	**30.0**	35.0	68.0	29.3 ± 8.9
Genotype quality GQ	1,273	5.0	99.0	**99.0**	99.0	99.0	91.4 ± 17.5
Heterozygous VAF	865	0.125	0.429	**0.500**	0.568	0.857	0.497 ± 0.105
Indel length (bp)	313	1.0	1.0	**1.0**	2.0	154.0	3.5 ± 10.1

QUAL and GQ. QUAL is the phred-scaled confidence that a variant exists (QUAL 30 = 1/1,000 error probability; clinical threshold QUAL ≥ 30). The median QUAL of 496 and minimum 31.6 show that no call falls below the threshold (0 calls below 30). GQ is the confidence in the genotype itself, capped at 99: 99 at the median and first quartile, i.e. most genotypes are certain; the low-GQ tail corresponds to the low-depth calls.

Figure 34. Read depth (DP) with the 15× clinical threshold.

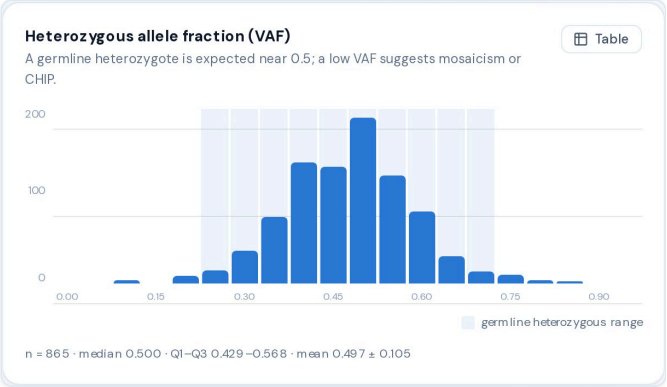


Figure 35. Heterozygous allele fraction (VAF).

Read depth. The distribution is centred on 30× (Q1–Q3 24–35×), exactly what a 30× genome gives at the variant sites. 90 calls (7.1 %) lie left of the 15× line — local coverage dips (GC-rich exons, repeats) — and are flagged DP<15: if one of them were pathogenic it would be reported "to confirm".

Heterozygous VAF. Germline heterozygous variants should carry ≈ 50 % of the reads. The median is 0.50 (Q1–Q3 0.43–0.57), a symmetric peak consistent with a single uncontaminated individual. The left tail holds 11 calls flagged LOW_VAF_MOSAIC_OR_CHIP (VAF < 0.25) — among them TP53 at 0.139 — and the right tail 7 calls above the germline range. The CHEK2 deletion has VAF 0.389 at DP 36: a clean heterozygote (indels are often slightly under-represented in reads).

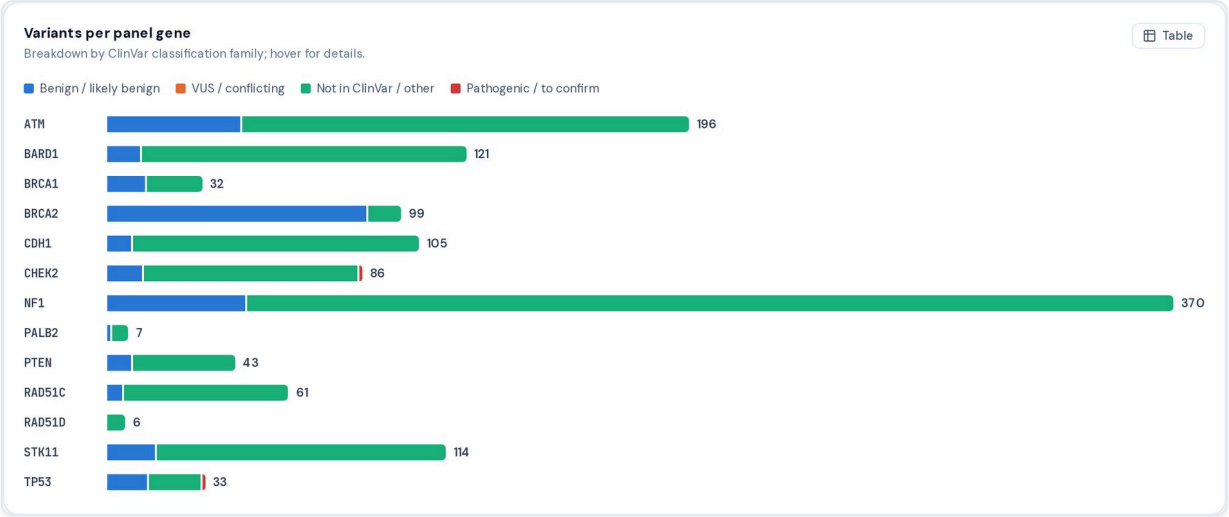


Figure 36. Variants per panel gene.

Table 21. NA10842 per gene: variant counts and coverage of known pathogenic sites.

Gene	Variants	Benign/LB	VUS	P/LP	P/LP sites ≥15×	Fraction	
NF1	370	48	0	0	5,541/5,548	99.9 %	
ATM	196	45	0	0	3,797/3,798	100.0 %	
BARD1	121	11	0	0	617/617	100.0 %	
STK11	114	16	0	0	389/389	100.0 %	
CDH1	105	8	0	0	618/618	100.0 %	
BRCA2	99	88	0	0	5,632/5,632	100.0 %	
CHEK2	86	12	0	1	960/1,022	93.9 %	**not excluded**
RAD51C	61	5	0	0	323/323	100.0 %	

Gene	Variants	Benign/LB	VUS	P/LP	P/LP sites ≥15×	Fraction
PTEN	43	8	0	0	1,312/1,312	100.0 %
TP53	33	14	0	1	1,142/1,142	100.0 %
BRCA1	32	13	0	0	4,013/4,013	100.0 %
PALB2	7	1	0	0	1,437/1,437	100.0 %
RAD51D	6	0	0	0	239/239	100.0 %

Variants per gene. Counts follow gene length and intronic content: large genes (NF1, ATM, BRCA2) carry the most variants, compact genes (TP53, CDH1, STK11) the fewest. They are not a risk measure by themselves.

NA10842 — quality checks and coverage

Table 22. Expert quality checks of NA10842.

Check	Value	Expected	Status	Interpretation
Transition/transversion ratio (PASS SNVs)	2.0420	1.8–3.3 (≥ 30 SNVs)	OK	A low Ti/Tv indicates false positives (random errors ≈ 0.5); ≈ 2.0 genome-wide, ≈ 3.0 coding exome.
Heterozygous/homozygous ratio	2.1200	1.0–3.0 (≥ 30 variants)	OK	Too high: possible contamination; too low: consanguinity or loss of heterozygosity.
Median heterozygous VAF	0.5000	0.4–0.6 (≥ 10 heterozygotes)	OK	A germline heterozygote is expected near 0.5; a shift suggests contamination or capture bias.
Share of PASS variants (GATK filters)	0.9810	≥ 0.8	OK	Many filtered variants: degraded sequencing or alignment quality.
Median depth at variants	30.0000	≥ 15 (clinical threshold)	OK	Insufficient depth: reduced sensitivity, less reliable genotypes.
Reads mapped to hg38	0.9991	≥ 0.95	OK	A low rate suggests contamination, a wrong reference or untrimmed adapters.
Duplicate rate (MarkDuplicates)	0.0720	≤ 0.30	OK	High rate: low library complexity, overestimated useful depth.
Panel ClinVar P/LP sites covered ≥ 15x	0.9973	≥ 0.95	OK	A poorly covered known pathogenic site cannot be excluded: no variant there does not mean no mutation.

Coverage of known pathogenic sites. 26,020 of 26,090 ClinVar P/LP sites (99.73 %) reach 15×, median 32×. Globally the analysis is conclusive (≥ 90 %), but CHEK2 reaches only 960/1,022 sites (93.9 %): CHEK2 has highly similar pseudogene copies where reads map with low quality and are excluded by the MAPQ ≥ 20 filter. A negative result in the uncovered part of CHEK2 would not exclude a mutation; the system states this explicitly ("not excluded") instead of hiding it.

NA12878 exome — how the same charts read on capture data

Table 23. Key values of the NA12878 exome.

Quantity	Value	Quantity	Value
Reads (exome)	90,115,545	Mapped / duplicates	99.81 % / 10.6 %
Panel variants	192	PASS	165 (85.9 %)
Ti/Tv (PASS SNVs)	2.41	Het/hom	1.91

Quantity	Value	Quantity	Value
Median depth at variants	19.5× (Q1–Q3 6.0–39.5)	Median heterozygous VAF	0.487
P/LP sites ≥ 15×	25,573/26,090 (98.0 %)	Genes not excluded	STK11 (90.8 %), TP53 (92.2 %)

Figure 37. NA12878 exome — read depth: bimodal shape.

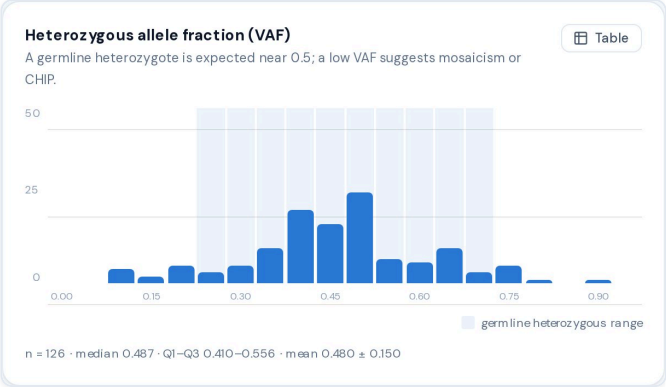


Figure 38. NA12878 exome — heterozygous VAF.

A bimodal depth distribution is the signature of exome capture. HaplotypeCaller is run over whole gene bodies (±100 bp), but an exome kit only captures coding exons: 65 of 192 calls (34 %) have DP < 10 — off-target intronic reads — while the on-target mode sits at 30–50×. The median (19.5×) therefore understates on-target depth. The 80 calls below 15× are flagged and could never produce a confirmed pathogenic finding. The PASS rate (85.9 %) is lower than in the genome for the same reason, and Ti/Tv is higher (2.41 vs 2.04) because coding sequence is enriched in transitions (CpG deamination).

Coverage gate in action. 98.0 % of the known pathogenic sites are covered at 15×, so the LOW conclusion is valid, but STK11 and TP53 fall below 95 % (353/389, 1,053/1,142): the report names them “not excluded”, and the BioGPT interpretation states it in plain language (Figure below).

Interpretation of the VCF statistics

Final text verified

Written by BioGPT fine-tuned on VCF statistics. Every sentence was checked against the computed statistics (numbers, expected ranges, risk level, genes); topics without a verified sentence use the deterministic reference text.

Generated sentences	Verified	Rejected	Topics from the model
13	13	0	13/13

BioGPT·verified	VARIANT CALLS	192 variants were called in the panel genes: 157 SNVs and 35 indels.
BioGPT·verified	GATK FILTERS	85.9% of the calls passed the GATK filters, as expected for good-quality data.
BioGPT·verified	TI/TV	The transition/transversion ratio (2.41) is within the expected range.
BioGPT·verified	HET/HOM	The het/hom ratio (1.91) is within the expected range.
BioGPT·verified	HETEROZYGOUS VAF	The median heterozygous VAF (0.49) is within the expected range.
BioGPT·verified	DEPTH	Variant sites have a median depth of 19.5x, adequate for clinical calling.
BioGPT·verified	MAPPING	99.8% of reads mapped to hg38, within the expected range (≥ 95%).
BioGPT·verified	DUPLICATES	Duplicates account for 10.6% of read pairs, which is acceptable.
BioGPT·verified	PATHOGENIC-SITE COVERAGE	98.0% of known pathogenic sites are covered at 15x or more, as expected.
BioGPT·verified	NOT EXCLUDED	Coverage is incomplete for STK11 and TP53: an uncovered known pathogenic variant cannot be excluded in this region.
BioGPT·verified	CONFIRMED P/LP	No confirmed pathogenic or likely pathogenic variant was found in the panel.
BioGPT·verified	LOW-VAF CALLS	11 calls have a low allele fraction (possible mosaicism or clonal haematopoiesis) and are flagged for review.
BioGPT·verified	RISK LEVEL	Hereditary breast cancer risk: LOW (deterministic rules).

Verification details (0 rejected sentences)

Figure 39. NA12878 — interpretation of the statistics: 13/13 sentences written by the fine-tuned BioGPT and verified, including the incomplete coverage of STK11 and TP53.

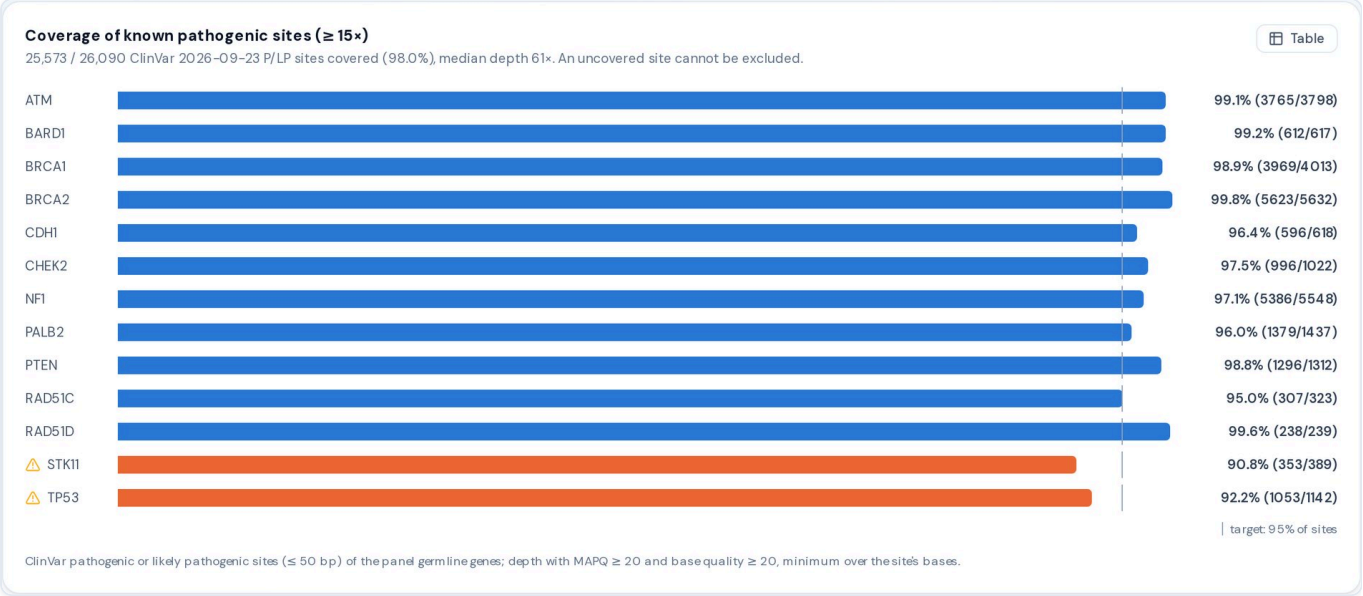


Figure 40. NA12878 exome — coverage of known pathogenic sites per gene.

Cohort-wide quality control

Table 24. Quality metrics of every validation sample (NA12878 = exome; others = 30× genomes restricted to the panel genes).

Sample	Variants	PASS	Ti/Tv	Het/hom	Het VAF	Depth	Mapped	Dup.	Sites ≥15×	Risk
NA19130	1,853	98.0 %	2.22	2.14	0.50	31.0×	99.90 %	10.2 %	99.98 %	**HIGH**
HG00611	1,553	98.3 %	1.95	3.88 **WARN**	0.49	34.0×	99.90 %	7.2 %	100.00 %	**HIGH**
HG02620	1,599	98.6 %	2.09	2.29	0.50	31.0×	99.88 %	10.4 %	99.97 %	**HIGH**
NA10842	1,273	98.1 %	2.04	2.12	0.50	30.0×	99.91 %	7.2 %	99.73 %	**MODERATE**
HG00596	1,332	98.4 %	2.16	2.06	0.49	41.0×	99.92 %	8.6 %	100.00 %	**MODERATE**
HG00096	1,634	99.0 %	2.14	1.54	0.50	32.0×	99.92 %	9.8 %	99.99 %	**LOW**
HG01112	1,395	98.2 %	2.08	1.69	0.50	32.0×	99.91 %	8.6 %	99.98 %	**LOW**
NA12878	192	85.9 %	2.41	1.91	0.49	19.5×	99.81 %	10.6 %	98.02 %	**LOW**

All genomes show consistent metrics: Ti/Tv 1.95–2.22, median heterozygous VAF 0.49–0.50, median depth 30×–41×, mapping ≥ 99.88 %, duplicates 7.2 %–10.4 %, site coverage ≥ 99.73 %. One value is flagged: **HG00611 has a het/hom ratio of 3.88**, above the 1.0–3.0 range. The system reports it for review without changing the risk. Its median heterozygous VAF (0.49) is centred, which argues against contamination (contamination shifts and widens the VAF peak); a panel-level ratio over ≈1,500 variants is sensitive to local haplotype structure. This is precisely the kind of case the expert checks exist for: flag, explain, let a human decide.

8. Biological interpretation of the findings

The validation cohort was chosen to contain real carriers of the most important mechanisms: two BRCA2 truncating variants, one PALB2 frameshift, the CHEK2 founder frameshift and an ATM splice-site variant, in individuals of African, East Asian and European ancestry, plus two non-carriers. The table gives the facts measured by GermlineIQ; the paragraphs explain what they mean biologically.

Table 25. Germline pathogenic variants of the cohort. Consequence and review status from the local ClinVar release; VAF and depth from the GermlineIQ reports.

Sample (population · sex)	Gene	Variant (GRCh38) · dbSNP · ClinVar	Consequence	ClinVar review	VAF (alt/depth)	Risk
NA19130 YRI · male	**BRCA2**	`NC_000013.11:g.32339596_32339597insTA` rs749980674 · VCV000254555	nonsense	reviewed by expert panel	0.42 (15/36)	**HIGH**
HG00611 CHS · female	**BRCA2**	`NC_000013.11:g.32357821dup` rs397507934 · VCV000052390	frameshift	reviewed by expert panel	0.41 (19/46)	**HIGH**
HG02620 GWD · male	**PALB2**	`NC_000016.10:g.23636314CA[1]` rs776284609 · VCV000949556	frameshift	multiple submitters, no conflicts	0.68 (13/19)	**HIGH**
NA10842 CEU · male	**CHEK2**	`NC_000022.11:g.28695869del` rs555607708 · VCV000128042	frameshift	multiple submitters, no conflicts	0.39 (14/36)	**MODERATE**
HG00596 CHS · female	**ATM**	`NC_000011.10:g.108271407G>T` rs192810283 · VCV000936903	splice donor	multiple submitters, no conflicts	0.51 (22/43)	**MODERATE**

BRCA2 — NA19130 (Yoruba, male) and HG00611 (Southern Han Chinese, female). Both variants truncate BRCA2. In NA19130 the insertion of two bases (TA) creates an immediate stop codon (ClinVar: nonsense); in HG00611 the duplication of one base shifts the reading frame. Both are classified pathogenic by the ClinVar expert panel. The mRNA is either degraded by nonsense-mediated decay or translated into a truncated protein lacking the C-terminal nuclear-localisation signals (and, depending on the position, part of the DNA-binding domain): BRCA2 cannot reach the break and load RAD51, and after loss of the second allele the breast cell cannot repair double-strand breaks by HR. Both are heterozygous (VAF 0.42 and 0.41), as expected for a germline variant. For the male carrier the same variant also increases the risk of male breast and prostate cancer and is transmitted to half of his children; GermlineIQ reports HIGH risk for both.

PALB2 — HG02620 (Gambian, male). The deletion shifts the reading frame of PALB2, whose protein bridges BRCA1 and BRCA2; without it BRCA2 is not recruited to the break. PALB2 is now considered a high-penetrance gene (OR ≈ 5) [Antoniou 2014; BCAC 2021]. The VAF of 0.68 at only 19× (13 reads) looks high, but its 95 % confidence interval (Figure 41) includes 0.5: at low depth a true heterozygote easily reaches this value, which is why the clinical QC window is 0.25–0.75 rather than 0.4–0.6.

CHEK2 c.1100delC — NA10842 (CEU, male). The deletion of one cytosine (p.Thr367MetfsTer15) truncates the checkpoint kinase and abolishes its activity; it is the classic founder variant of Northern and Eastern European populations and doubles breast cancer risk [Meijers-Heijboer 2002; CHEK2 consortium 2004]. NA10842 belongs to a CEPH/Utah family of Northern European ancestry — consistent with a founder variant. Because the 1000 Genomes collection also contains his parents, the transmission can be checked on real data: the 1000 Genomes call set gives `NA10842 = 1|0`, father `NA11917 = 0|1`, mother `NA11918 = 0|0`. The variant was **inherited from the father**, exactly as autosomal dominant transmission predicts. GermlineIQ reports MODERATE risk, names CHEK2 “not excluded” for the part of the gene its reads could not cover (CHEK2 has pseudogene copies that attract ambiguous reads), and its verifier rejected BioGPT's claim of a “high” risk.

ATM — HG00596 (Southern Han Chinese, female). The variant alters a splice donor site: the exon is skipped or the intron retained, giving a non-functional ATM kinase. Heterozygous carriers have a moderately increased breast cancer risk; biallelic carriers have ataxia-telangiectasia [Renwick 2006]. VAF 0.51: a textbook heterozygote. Risk: MODERATE.

Controls — HG00096 (British) and HG01112 (Colombian). No P/LP variant, and ≥ 99.98 % of the known pathogenic sites covered at 15×: the negative result is informative, and GermlineIQ concludes LOW. The heterozygosity of the samples follows population history — het/hom ratio 2.14 and 2.29 in the two African-ancestry genomes versus 1.54 in the British genome — as expected from the greater genetic diversity of African populations.

The TP53 p.Arg273His call: germline or acquired?

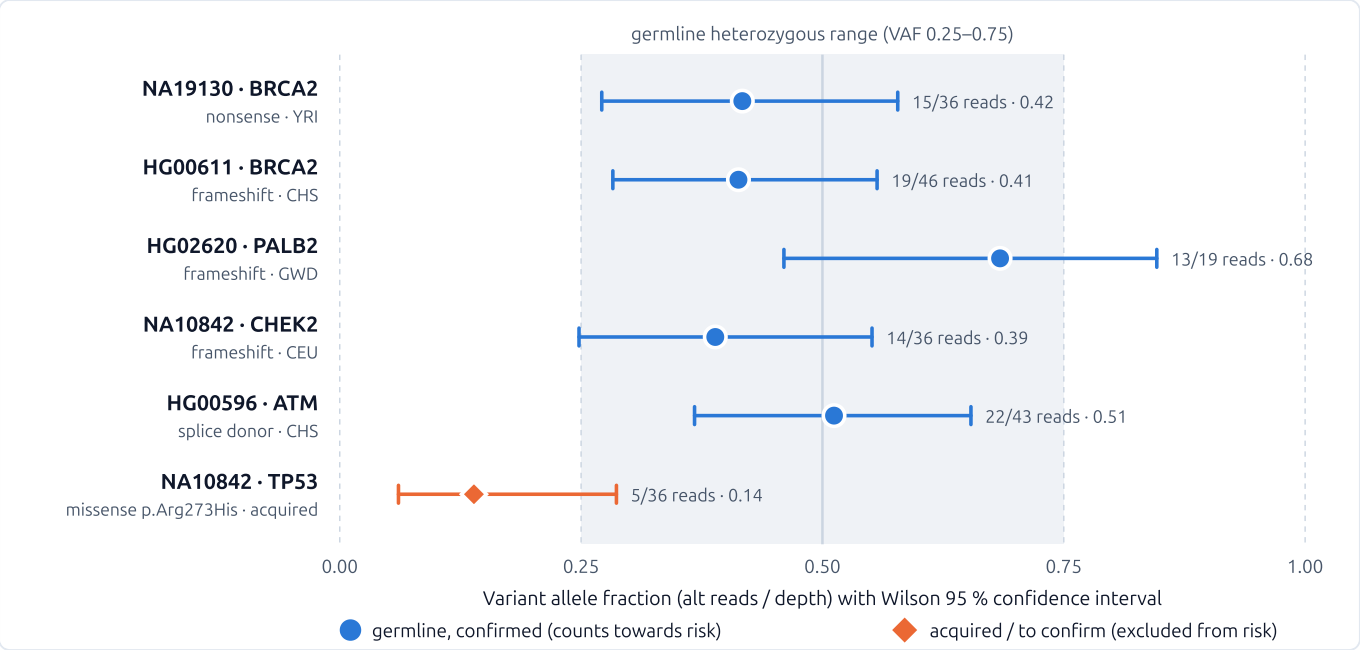


Figure 41. VAF of every pathogenic call found in the cohort, with its Wilson 95 % confidence interval (binomial sampling of reads). The five germline variants lie in the heterozygous band and their intervals include 0.5; the TP53 call (5/36 reads, 95 % CI 0.06–0.29) is statistically incompatible with a heterozygous germline variant.

In NA10842, HaplotypeCaller also found TP53 c.818G>A, **p.Arg273His** (rs28934576), classified pathogenic by the ClinVar expert panel. Arg273 is one of the residues through which p53 contacts DNA [Cho 1994]; its substitution abolishes p53's binding to its target genes and it is one of the most frequent mutations in human cancers [Olivier 2010]. If this variant were germline, the person would have Li-Fraumeni syndrome and the report should say HIGH. Three independent lines of evidence show that it is **not inherited**:

- **Allele fraction** — 5 of 36 reads (VAF 0.14); the 95 % interval 0.06–0.29 excludes 0.5 (Figure 41). Only a fraction of the cells carry it.
- **Population call set** — the variant is absent from NA10842's germline genotype in the 1000 Genomes high-coverage call set.
- **Family** — neither parent (NA11917, NA11918) carries it: it is not transmitted, and at this VAF it is not a constitutional de-novo variant either.

1000 Genomes DNA comes from lymphoblastoid cell lines (EBV-immortalised B cells); a TP53 hotspot mutation at low fraction is typical of a clone that expanded in culture, or of clonal haematopoiesis in the donor's blood, where TP53 is one of the recurrently mutated genes [Jaiswal 2014; Genovese 2014]. GermlineIQ reached the right conclusion from the reads alone, before any of this external evidence was consulted: its VAF-aware QC classified the call “to confirm” (LOW_VAF_MOSAIC_OR_CHIP, plus the GATK QD filter) and excluded it from the risk, which stayed MODERATE (CHEK2). **This is the key biological result of the validation:** a pipeline that treats every ClinVar-pathogenic call as germline would have produced a false Li-Fraumeni diagnosis.

Clinical translation. For a real patient, the report's “to confirm” would lead the laboratory to re-test on a second tissue (e.g. skin fibroblasts or a new blood sample) with an orthogonal method, and the genetic counsellor to consider clonal haematopoiesis. The CHEK2 and BRCA2 findings, in contrast, are germline and open cascade testing of relatives — which the NA10842 trio illustrates.

All validation samples as written in their reports

Table 26. Clinical findings of the validation samples as written in their reports.

Sample	Findings (ClinVar, GATK metrics)	Risk
NA19130	**BRCA2** `NC_000013.11:g.32339596_32339597insTA` (rs749980674) Pathogenic 3★, heterozygous, VAF 0.417, DP 36	**HIGH**
HG00611	**BRCA2** `NC_000013.11:g.32357821dup` (rs397507934) Pathogenic 3★, heterozygous, VAF 0.413, DP 46	**HIGH**
HG02620	**PALB2** `NC_000016.10:g.23636314CA[1]` (rs776284609) Pathogenic 2★, heterozygous, VAF 0.684, DP 19	**HIGH**
NA10842	**CHEK2** `NC_000022.11:g.28695869del` (rs555607708) Pathogenic 2★, heterozygous, VAF 0.389, DP 36 *to confirm:* TP53 `NC_000017.11:g.7673802C>T` VAF 0.139 (FILTER:SNP_QD2, LOW_VAF_MOSAIC_OR_CHIP)	**MODERATE**
HG00596	**ATM** `NC_000011.10:g.108271407G>T` (rs192810283) Pathogenic 2★, heterozygous, VAF 0.512, DP 43	**MODERATE**
HG00096 (control)	No P/LP variant	**LOW**
HG01112 (control)	No P/LP variant	**LOW**
NA12878 (GIAB exome)	No P/LP variant	**LOW**

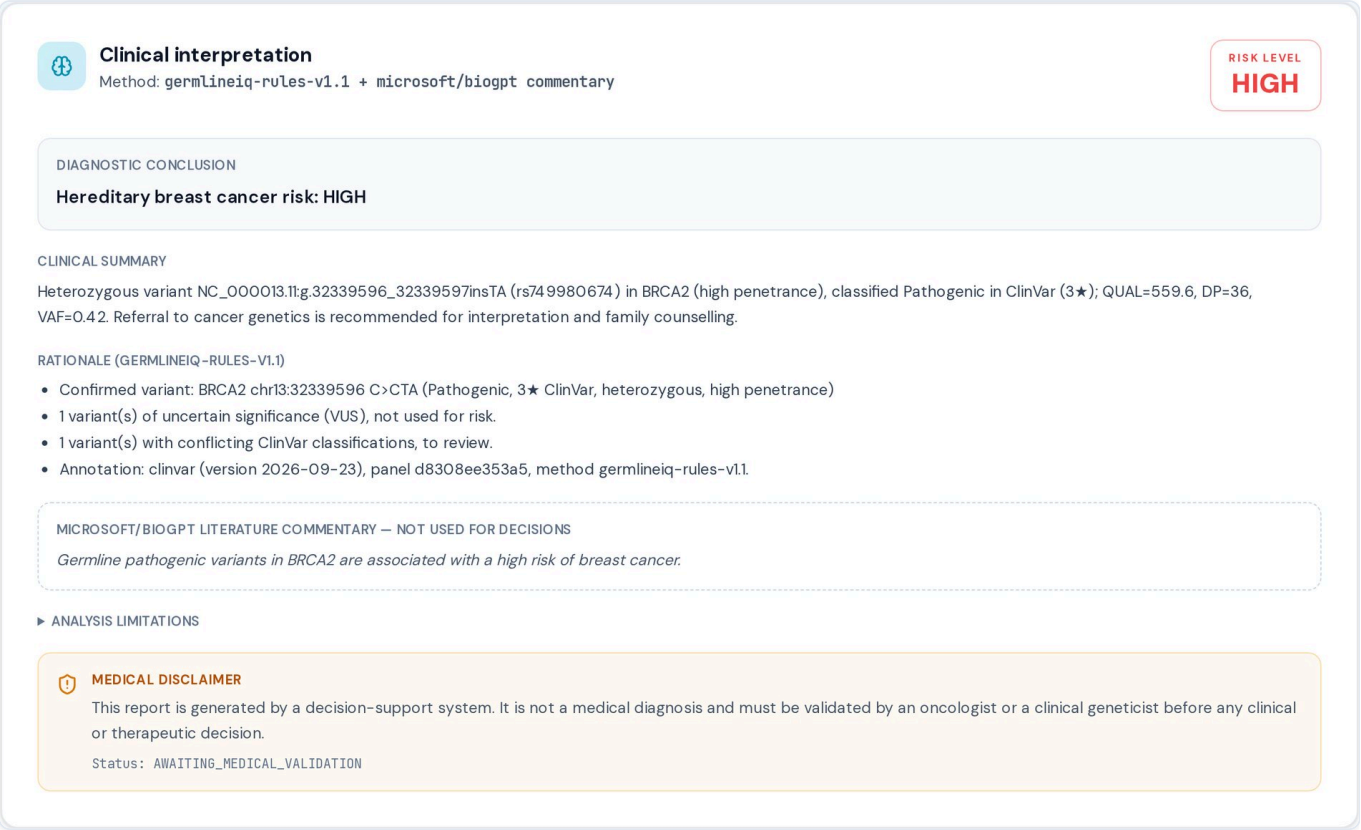


Figure 42. NA19130 — HIGH risk: confirmed BRCA2 2-bp insertion creating a stop codon (ClinVar: nonsense; Pathogenic, expert panel).

Interpretation of the VCF statistics

Final text verified

Written by BioGPT fine-tuned on VCF statistics. Every sentence was checked against the computed statistics (numbers, expected ranges, risk level, genes); topics without a verified sentence use the deterministic reference text.

Generated sentences	Verified	Rejected	Topics from the model
12	12	0	12/12

BioGPT·verified	VARIANT CALLS	1853 variants were called in the panel genes: 1407 SNVs and 446 indels.
BioGPT·verified	GATK FILTERS	98.0% of the calls passed the GATK filters, as expected for good-quality data.
BioGPT·verified	TI/TV	The transition/transversion ratio (2.22) is within the expected range.
BioGPT·verified	HET/HOM	The het/hom ratio (2.14) is within the expected range.
BioGPT·verified	HETEROZYGOUS VAF	The median heterozygous VAF (0.50) is within the expected range.
BioGPT·verified	DEPTH	Variant sites have a median depth of 31x, adequate for clinical calling.
BioGPT·verified	MAPPING	99.9% of reads mapped to hg38, within the expected range (≥ 95%).
BioGPT·verified	DUPLICATES	Duplicates account for 10.2% of read pairs, which is acceptable.
BioGPT·verified	PATHOGENIC-SITE COVERAGE	100.0% of known pathogenic sites in the panel are covered at 15x or more, within the expected range (≥ 95%).
BioGPT·verified	CONFIRMED P/LP	The analysis confirmed a pathogenic or likely pathogenic variant in BRCA2.
BioGPT·verified	LOW-VAF CALLS	11 calls have a low allele fraction (possible mosaicism or clonal haematopoiesis) and are flagged for review.
BioGPT·verified	RISK LEVEL	Hereditary breast cancer risk: HIGH (deterministic rules).

Verification details (0 rejected sentences)

Figure 43. NA19130 — verified interpretation of the statistics.

9. Research contributions — added value of the project

GermlineIQ is a research platform: beyond integrating standard tools (BWA, GATK, Parabricks, ClinVar), it proposes and evaluates mechanisms that address known weaknesses of automated genomic reporting — hallucinating language models, false reassurance from incomplete coverage, somatic events mistaken for germline variants, and irreproducible multi-agent systems.

Table 27. Scientific and technical contributions, each with its evidence.

#	Contribution	What is new	Evidence in this work
1	**Verifier-in-the-loop generation** for genomic reports	A language model is fine-tuned to verbalise VCF statistics, and a deterministic verifier checks each sentence (numbers against the computed values, status words against the thresholds, genes against their role, risk against the rules) before it can enter the report; unverified topics fall back to reference text, so the final text is correct by construction. Promotion is gated by measured factuality, not by perplexity.	Base BioGPT 0 % verified → fine-tuned 98.6 % (synthetic) and 100 % (15 real analyses); 0 wrong risk levels; final text verified in 100 % of reports.
2	**Penetrance-consistent knowledge verification**	Literature sentences are checked not only for gene–disease validity (ClinGen/NCCN) but also for the magnitude of risk against the gene's penetrance class.	BioGPT's “CHEK2 … high risk” rejected and replaced in the real NA10842 report.
3	**Coverage-aware decision rules**	The risk decision uses the depth at each of the 26,090 known pathogenic sites of the panel, so that a “low risk” is only issued when the absence of a variant is interpretable; insufficiently covered genes are named.	CHEK2 (93.9 %) and STK11/TP53 in the exome flagged “not excluded”; negative results of controls supported by ≥ 99.98 % coverage.
4	**VAF-aware germline/somatic triage**	Allele fraction and GATK filters decide whether a pathogenic call can support a germline conclusion.	TP53 p.Arg273His correctly excluded; its acquired nature confirmed a posteriori by the trio and the population call set.
5	**Deterministic multi-agent orchestration with an LLM router**	An LLM may choose the order of ready tools but never the content; clinical content is hashed (SHA-256) to prove equivalence.	Deterministic and Mistral routers → identical plan and byte-identical report; reproducible over repeated runs.
6	**N-version independent verification**	A second implementation (bcftools/samtools/awk, no shared code) recomputes 20 metrics of every report.	20/20 on 8/8 validation samples; the tool even exposed and fixed a cache-traceability issue.
7	**Open, privacy-free validation strategy**	Real positive controls are discovered by scanning 3,202 public genomes for ClinVar P/LP variants, then only the panel reads are streamed from the public CRAM files.	Five real carriers and two controls, analysed end to end without any patient data.
8	**Local, GPU-accelerated, fail-safe deployment**	Parabricks on a consumer Blackwell GPU with automatic GATK fallback and service-user containers; data never leave the server.	GPU and CPU final VCFs identical (1,357/1,357); exome falls back automatically when host RAM is insufficient.
9	**A documented negative result**	Domain fine-tuning of BioGPT on PubMed lowered perplexity (15.8 → 8.6) without improving factual accuracy on held-out prompts.	Four iterations, none promoted: perplexity is not a proxy for factuality in clinical text.

Comparison with a conventional workflow

Aspect	Conventional pipeline + manual interpretation	GermlineIQ
Variant calling	GATK best practices on CPU (hours per exome)	Same algorithms, GPU-accelerated, automatic CPU fallback, panel-restricted
Classification	Manual ClinVar look-ups, version often unrecorded	Exact-allele ClinVar matching, release recorded in each report

Aspect	Conventional pipeline + manual interpretation	GermlineIQ
Negative result	Usually reported without site-level coverage	Coverage of every known pathogenic site; genes “not excluded” named
Somatic / mosaic calls	Risk of being reported as germline	VAF-aware triage, reported “to confirm”
Narrative text	Written by hand, or by an unverified language model	Written by a fine-tuned model, verified sentence by sentence
Traceability	Scattered files and logs	One JSON artifact per agent, SHA-256, versions, independent recomputation
Data location	Often cloud platforms	Local server only

Perspectives

The same architecture extends naturally to copy-number variants (large BRCA1 rearrangements), to other hereditary cancer panels (Lynch syndrome), to automatic ACMG/AMP evidence scoring, and to larger validation sets (GIAB HG002–HG007, GeT-RM reference materials). The verifier-in-the-loop principle is general: any clinical text produced by a language model can be constrained to facts computed by deterministic code.

10. Limitations and roadmap

- **Variant types.** SNVs and small indels only; large rearrangements and copy-number variants (e.g. BRCA1 exon deletions) are not detected — stated in every report. *Roadmap:* CNV calling on the panel.
- **Classification.** Pathogenicity comes from ClinVar (exact allele, versioned release); novel variants are reported as not in ClinVar or, for predicted loss of function, “to confirm”. No automatic ACMG/AMP scoring.
- **Validation size.** Seven real genomes and one GIAB exome: confidence intervals are wide (risk accuracy 95 % CI 64.6–100 %). *Roadmap:* GIAB HG002–HG007, GeT-RM positive controls, stratified hap.py metrics.
- **Hard filters.** GATK default hard filters cost recall on exome data (SOR); kept by design, with filtered pathogenic variants reported “to confirm”.
- **GPU memory on the validation server.** Parabricks needs more host RAM than the 18 GB assigned to WSL2 for an exome; the pipeline falls back automatically to GATK4 on CPU (same calls, ≈1 h per exome).
- **BioGPT.** The statistics interpreter is trained on rule-based reference texts: it verbalises, it does not discover. The literature-commentary adapter did not pass its gate and is not used.
- **Regulatory status.** Research software; clinical use requires medical validation and a regulatory pathway (IVDR in the EU).

11. References

1. McKenna A, Hanna M, Banks E, et al. The Genome Analysis Toolkit: a MapReduce framework for analyzing next-generation DNA sequencing data. *Genome Res.* 2010;20(9):1297–1303.
2. DePristo MA, Banks E, Poplin R, et al. A framework for variation discovery and genotyping using next-generation DNA sequencing data. *Nat Genet.* 2011;43(5):491–498.
3. Van der Auwera GA, Carneiro MO, Hartl C, et al. From FastQ data to high confidence variant calls: the Genome Analysis Toolkit best practices pipeline. *Curr Protoc Bioinformatics.* 2013;43:11.10.1–11.10.33.
4. Poplin R, Ruano-Rubio V, DePristo MA, et al. Scaling accurate genetic variant discovery to tens of thousands of samples. *bioRxiv* 2017:201178.
5. Li H. Aligning sequence reads, clone sequences and assembly contigs with BWA-MEM. *arXiv*:1303.3997, 2013.
6. Danecek P, Bonfield JK, Liddle J, et al. Twelve years of SAMtools and BCFtools. *GigaScience.* 2021;10(2):giab008.
7. NVIDIA. Clara Parabricks documentation, version 4.6 (fq2bam, haplotypecaller). docs.nvidia.com/clara/parabricks.
8. Broad Institute. Picard toolkit (MarkDuplicates metrics). broadinstitute.github.io/picard.
9. Landrum MJ, Lee JM, Benson M, et al. ClinVar: improving access to variant interpretations and supporting evidence. *Nucleic Acids Res.* 2018;46(D1):D1062–D1067.
10. Richards S, Aziz N, Bale S, et al. Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the ACMG and AMP. *Genet Med.* 2015;17(5):405–424.
11. Strande NT, Riggs ER, Buchanan AH, et al. Evaluating the clinical validity of gene–disease associations: an evidence-based framework developed by the Clinical Genome Resource. *Am J Hum Genet.* 2017;100(6):895–906.
12. National Comprehensive Cancer Network. NCCN Clinical Practice Guidelines in Oncology: Genetic/Familial High-Risk Assessment — Breast, Ovarian, Pancreatic, and Prostate (current version).
13. Breast Cancer Association Consortium; Dorling L, Carvalho S, Allen J, et al. Breast cancer risk genes — association analysis in more than 113,000 women. *N Engl J Med.* 2021;384(5):428–439.
14. Hu C, Hart SN, Gnanaolivu R, et al. A population-based study of genes previously implicated in breast cancer. *N Engl J Med.* 2021;384(5):440–451.
15. Couch FJ, Shimelis H, Hu C, et al. Associations between cancer predisposition testing panel genes and breast cancer. *JAMA Oncol.* 2017;3(9):1190–1196.
16. Jaiswal S, Fontanillas P, Flannick J, et al. Age-related clonal hematopoiesis associated with adverse outcomes. *N Engl J Med.* 2014;371(26):2488–2498.
17. Zook JM, Chapman B, Wang J, et al. Integrating human sequence data sets provides a resource of benchmark SNP and indel genotype calls. *Nat Biotechnol.* 2014;32(3):246–251.
18. Wagner J, Olson ND, Harris L, et al. Benchmarking challenging small variants with linked and long reads. *Cell Genomics.* 2022;2(5):100128.
19. Krusche P, Trigg L, Boutros PC, et al. Best practices for benchmarking germline small-variant calls in human genomes. *Nat Biotechnol.* 2019;37(5):555–560.
20. Byrska-Bishop M, Evani US, Zhao X, et al. High-coverage whole-genome sequencing of the expanded 1000 Genomes Project cohort including 602 trios. *Cell.* 2022;185(18):3426–3440.
21. Luo R, Sun L, Xia Y, et al. BioGPT: generative pre-trained transformer for biomedical text generation and mining. *Brief Bioinform.* 2022;23(6):bbac409.
22. Hu EJ, Shen Y, Wallis P, et al. LoRA: Low-Rank Adaptation of Large Language Models. *ICLR* 2022 (arXiv:2106.09685).
23. Jiang AQ, Sablayrolles A, Mensch A, et al. Mistral 7B. *arXiv*:2310.06825, 2023.
24. Wilson EB. Probable inference, the law of succession, and statistical inference. *J Am Stat Assoc.* 1927;22(158):209–212.
25. Hyndman RJ, Fan Y. Sample quantiles in statistical packages. *Am Stat.* 1996;50(4):361–365.
26. LangChain. LangGraph documentation (stateful multi-agent graphs). langchain-ai.github.io/langgraph.
27. Moynahan ME, Jasin M. Mitotic homologous recombination maintains genomic stability and suppresses tumorigenesis. *Nat Rev Mol Cell Biol.* 2010;11(3):196–207.

28. Xia B, Sheng Q, Nakanishi K, et al. Control of BRCA2 cellular and clinical functions by a nuclear partner, PALB2. *Mol Cell.* 2006;22(6):719–729.

29. Knudson AG. Mutation and cancer: statistical study of retinoblastoma. *Proc Natl Acad Sci USA.* 1971;68(4):820–823.

30. Kuchenbaecker KB, Hopper JL, Barnes DR, et al. Risks of breast, ovarian, and contralateral breast cancer for BRCA1 and BRCA2 mutation carriers. *JAMA.* 2017;317(23):2402–2416.

31. Antoniou AC, Casadei S, Heikkinen T, et al. Breast-cancer risk in families with mutations in PALB2. *N Engl J Med.* 2014;371(6):497–506.

32. Malkin D, Li FP, Strong LC, et al. Germ line p53 mutations in a familial syndrome of breast cancer, sarcomas, and other neoplasms. *Science.* 1990;250(4985):1233–1238.

33. Meijers-Heijboer H, van den Ouweland A, Klijn J, et al. Low-penetrance susceptibility to breast cancer due to CHEK2*1100delC in noncarriers of BRCA1 or BRCA2 mutations. *Nat Genet.* 2002;31(1):55–59.

34. CHEK2 Breast Cancer Case-Control Consortium. CHEK2*1100delC and susceptibility to breast cancer: a collaborative analysis involving 10,860 breast cancer cases and 9,065 controls from 10 studies. *Am J Hum Genet.* 2004;74(6):1175–1182.

35. Renwick A, Thompson D, Seal S, et al. ATM mutations that cause ataxia-telangiectasia are breast cancer susceptibility alleles. *Nat Genet.* 2006;38(8):873–875.

36. Cho Y, Gorina S, Jeffrey PD, Pavletich NP. Crystal structure of a p53 tumor suppressor–DNA complex: understanding tumorigenic mutations. *Science.* 1994;265(5170):346–355.

37. Olivier M, Hollstein M, Hainaut P. TP53 mutations in human cancers: origins, consequences, and clinical use. *Cold Spring Harb Perspect Biol.* 2010;2(1):a001008.

38. Genovese G, Kähler AK, Handsaker RE, et al. Clonal hematopoiesis and blood-cancer risk inferred from blood DNA sequence. *N Engl J Med.* 2014;371(26):2477–2487.

39. Lykke-Andersen S, Jensen TH. Nonsense-mediated mRNA decay: an intricate machinery that shapes transcriptomes. *Nat Rev Mol Cell Biol.* 2015;16(11):665–677.

40. The 1000 Genomes Project Consortium. A global reference for human genetic variation. *Nature.* 2015;526(7571):68–74.

41. Farmer H, McCabe N, Lord CJ, et al. Targeting the DNA repair defect in BRCA mutant cells as a therapeutic strategy. *Nature.* 2005;434(7035):917–921.

12. Appendix — reproducibility and commands

Item	Value
Software	GermlineIQ 3.0.0, commit `83cc9a4`
Decision method	germlineiq-rules-v1.1
Statistics / alignment QC	germlineiq-stats-v1 / germlineiq-alnqc-v1
Gene–disease knowledge	germlineiq-gene-disease-v2 (penetrance-consistent)
Statistics interpreter	facts v1 · reference v2 · verifier v1 · adapter v1 promoted 2026-09-26
Interpreter dataset SHA-256 (train)	`1cb0cbc5ecc66e3bfa0e21c8c6292507…`
Panel	d8308ee353a5 (13 germline genes, ±100 bp)
ClinVar	GRCh38, release 2026-09-23
Benchmark	germlineiq-bench-v1 · started 2026-09-27T13:26:56 · pipeline parabricks

```
# Start / stop (systemd user service)
systemctl --user start germlineiq          # also: stop | restart | status
bash scripts/smoke_test.sh --fastq R1.fastq.gz R2.fastq.gz

# Independent verification of a report
python scripts/expert_check.py $LOCAL_DATA_ROOT/patients/<ID>/output

# Full benchmark (results shown in the Benchmarks page)
cd Backend && python -m benchmarks.multiagent --suites all

# BioGPT statistics interpreter: dataset → train → evaluate → promote (gate)
cd Backend && python -m src.llm.stats_finetune all

# Tests
cd Backend && python -m pytest -q          # 177 tests
```